

I. Hisoblashga doir mashqlar

1. Hisoblang: $\sqrt{2} \cdot \sqrt{8} - 1,25$ J: 2,75

2. Hisoblang: $\frac{\sqrt{200}}{\sqrt{2}} - 0,2$ J: 9,8

3. Hisoblang: $2,1 - \sqrt{0,036} \cdot \sqrt{10}$ J: 1,5

4. Hisoblang: $\sqrt{49} - 0,5^5 : 0,5^3$ J: 6,75

5. Hisoblang: $\sqrt{3} \cdot \sqrt{12} - 5,45$ J: 0,55

6. Hisoblang: $\frac{\sqrt{800}}{\sqrt{2}} + \sqrt{9}$ J: 23

7. Hisoblang: $3,2 - \sqrt{0,9} : \sqrt{10}$ J: 0,2

8. Hisoblang: $\sqrt{64} - 0,1^8 : 0,1^6$ J: 7,99

9. Hisoblang: $\sqrt{2} \cdot \sqrt{50} - 7,25$ J: 2,75

10. Hisoblang: $\frac{\sqrt{81} + 3\sqrt{4}}{\sqrt{100}}$ J: 1,5

11. Hisoblang: $4,3 - \sqrt{0,32} \cdot \sqrt{2}$ J: 3,5

12. Hisoblang: $\sqrt{16} + 0,3^{12} : 0,3^{10}$ J: 3,91

13. Hisoblang: $\sqrt{2} \cdot \sqrt{18} - 2,05$ J: 1,95

14. Hisoblang: $\sqrt{500} \cdot \sqrt{\frac{5}{4}}$ J: 25

15. Hisoblang: $7,3 - \sqrt{0,049} \cdot \sqrt{10}$ J: 6,6

16. Hisoblang: $\sqrt{3} \cdot \sqrt{27} - 0,5^6 : 0,5^5$ J: 8,5

II. Amaliy masalalar

1. Telefonning narxi 7 000 000 so'm. Aksiya vaqtida Sardor telefonni do'kondan 10 % chegirma bilan sotib oldi. Sardor telefon uchun qancha pul to'lagan?
J: 6 100 000 so'm
2. Amir 2 soat davomida 12 km/h tezlikda, so'ng 0,5 soat davomida 8 km/h tezlikda harakatlandi. U bosib o'tgan umumiy masofani toping.
J: 28 km
3. Zilola oilasi uchun osh tayyorlashda 600 g go'sht ishlatadi. Uning oilasida 5 kishi (Zilola, eri va uch nafar farzandi) bor. Ertasi kuni ularnikiga yana 15 nafar mehmon keladi. Zilola oshni ham oila a'zolari, ham mehmonlar uchun tayyorlaydi. U oshga jami necha gramm go'sht ishlatadi?
J: 2400 g
4. Sentabr oyida apelsinning 1 kg narxi 20 000 so'm edi. Oktabr oyida narx 20% ga oshdi. Oktabr oyida 1 kg apelsinning narxi qancha bo'ldi?
J: 24 000 so'm
5. «Yangilik» do'konida sumkaning narxi 400 000 so'm edi. Navro'z bayrami munosabati bilan barcha mahsulotlarga 20 % chegirma berildi. Chegirmadan keyingi sumkaning narxini toping.
J: 320 000 so'm
6. Alisher Toshkentdan Samarqandga 3 soat davomida 80 km/h tezlikda harakatlandi. Qaytishda shu yo'lni tirbandlik sababli u 4 soatda bosib o'tdi. Alisherning qaytishdagi o'rtacha tezligini toping.
J: 60 km/h
7. 8 kishiga sharbat tayyorlash uchun Feruzaga 2 litr suv va 400 g quruq meva kerak bo'ladi. Shu retsept asosida u ishlaydigan ofisdagi 20 kishi uchun sharbat tayyorlashi uchun necha gramm quruq meva kerak bo'ladi?
J: 1000 g
8. Fevral oyida Azizning oylik maoshi 5 000 000 so'mni tashkil etgan. Mart oyida muvaffaqiyatli loyiha sababli uning ish haqi 20 % ga oshirildi. Azizning yangi oylik maoshini toping.
J: 6 000 000 so'm
9. Ko'ylakning narxi 400 000 so'm. Aksiya vaqtida ko'ylakka 10% chegirma berildi. Chegirmadan keyingi ko'ylak narxini toping.
J: 320 000 so'm
10. Said avtomobilida 2 soat davomida o'rtacha 60 km/h tezlikda, so'ng 0,5 soat davomida o'rtacha 20 km/h tezlikda harakatlandi. Said bosib o'tgan umumiy masofani toping.
J: 130 km
11. 40 mm^2 maydonga plitka yotqizish uchun 8 qop plitka yelimi kerak bo'ladi. Shu turdagi yelimdan 100 mm^2 maydonga plitka yotqizish uchun necha qop kerak bo'ladi?
J: 20 qop

12. Mart oyining boshida 1 kg qulupnayning narxi 150 000 so'm edi. Bir oy o'tgach, qulupnay 40% ga arzonlashdi. Aprel oyining boshida 1 kg qulupnayning narxi qancha bo'ldi?
J: 90 000 so'm
13. Noutbukning narxi 9 000 000 so'm. Aksiya doirasida do'kon 10 % chegirma e'lon qildi. Shahzod noutbuk sotib olmoqchi. U chegirma bilan noutbuk uchun qancha pul to'laydi?
J: 8 100 000 so'm
14. Rustam shahar tashqarisidagi avtomobil yo'lida 2 soat davomida 90 km/h tezlikda harakatlandi. Qaytishda, xuddi shu yo'lda, tirbandlik vaqtida u 3 soat yo'l yurdi. Rustamning qaytishdagi o'rtacha tezligini toping.
J: 60 km/h
15. Umidga 6 metr uzunlikdagi panjarani bo'yash uchun 2 litr bo'yoq kerak bo'ladi. 21 metr uzunlikdagi shunday panjarani bo'yash uchun necha litr bo'yoq kerak bo'ladi?
J: 7 litr
16. «Chilonzor» sataroshxonasida soch olish xizmati narxi 50 000 so'm edi. Ta'mirlash va interyer yangilanganidan so'ng narx 30 % ga oshirildi. Endi soch olish xizmati qancha turadi?
J: 65 000 so'm

III. Algebraik ifodalar

1. $a^2 + 6a - (a + 2)^2$ ifodani soddalashtiring va $a = -3$ bo'lgandagi qiymatini toping.
J: -2
2. $\frac{2a^2-6a}{a-3} - 4a + 8$ ifodani soddalashtiring va $a = 4$ bo'lgandagi qiymatini toping.
J: 0
3. $(a - 3)^2 - a(a - 3)$ ifodani soddalashtiring va $a = -4$ bo'lgandagi qiymatini toping.
J: 21
4. $\frac{2a^3-4a^2}{a^2} - 7a + 3$ ifodani soddalashtiring va $a = 3$ bo'lgandagi qiymatini toping.
J: -16
5. $(a - 4)(a - 3) - a^2 + 2a$ ifodani soddalashtiring va $a = -2$ bo'lgandagi qiymatini toping.
J: 22
6. $\frac{a^2}{a-2} - \frac{4}{a-2} - 5a + 2$ ifodani soddalashtiring va $a = 3$ bo'lgandagi qiymatini toping.
J: -8
7. $(a - 5)(a + 5) - a(a - 3)$ ifodani soddalashtiring va $a = -5$ bo'lgandagi qiymatini toping.
J: -40
8. $\frac{a^2}{a+1} \cdot \frac{3a+3}{a} - a - 8$ ifodani soddalashtiring va $a = 3$ bo'lgandagi qiymatini toping.
J: -2

9. $a^2 + 4a - (a + 3)^2$ ifodani soddalashtiring va $a = -2$ bo'lgandagi qiymatini toping.

J: -5

10. $\frac{2a^2+8a}{a+4} - 5a + 2$ ifodani soddalashtiring va $a = 3$ bo'lgandagi qiymatini toping.

J: -13

11. $(a - 4)^2 - a(a - 6)$ ifodani soddalashtiring va $a = -3$ bo'lgandagi qiymatini toping.

J: 22

12. $\frac{2a^3+10a^2}{a^2} - 5a + 4$ ifodani soddalashtiring va $a = 5$ bo'lgandagi qiymatini toping.

J: -1

13. $(a - 5)(a + 3) - a^2 + 4a$ ifodani soddalashtiring va $a = -2$ bo'lgandagi qiymatini toping.

J: -19

14. $\frac{a^2}{a+3} - \frac{9}{a+3} - a + 5$ ifodani soddalashtiring va $a = 10$ bo'lgandagi qiymatini toping.

J: 2

15. $(a - 4)(a + 4) - a(a - 6)$ ifodani soddalashtiring va $a = -3$ bo'lgandagi qiymatini toping.

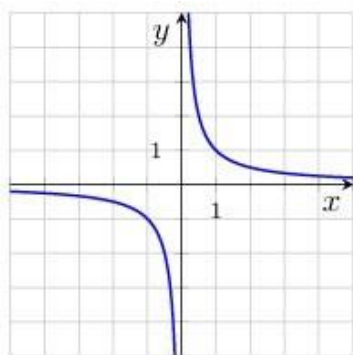
J: -34

16. $\frac{a^2}{a+5} \cdot \frac{2a+10}{a} - 4a + 5$ ifodani soddalashtiring va $a = 4$ bo'lgandagi qiymatini toping.

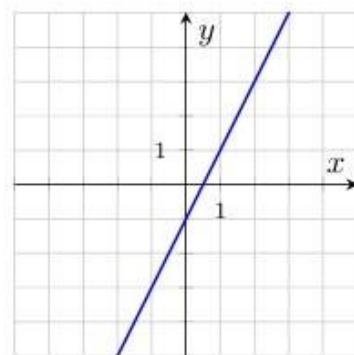
J: -3

IV Elementar funksiyalarning grafiklari

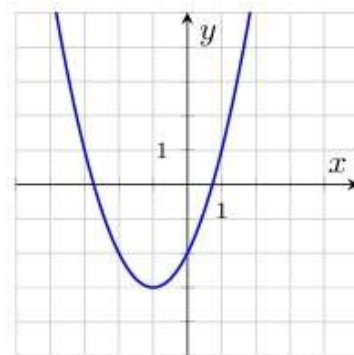
1. Quyida keltirilgan oltita funksiya uchta grafiklari rasmda tasvirlangan. Grafiklarga mos funksiyaning nomini toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = 2x - 1$

3) $y = x^2 + 2x - 2$

5) $y = -\frac{1}{x}$

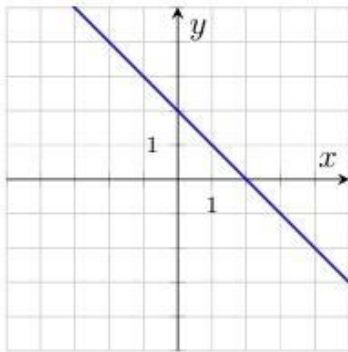
2) $y = -2x - 1$

4) $y = \frac{1}{x}$

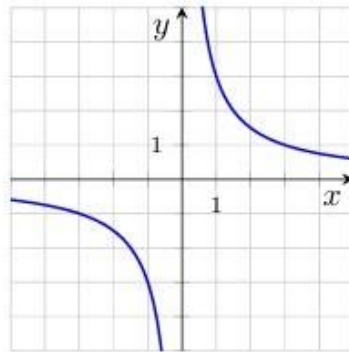
6) $y = -x^2 + 2x - 2$

	A	B	C
Javob:	4	1	3

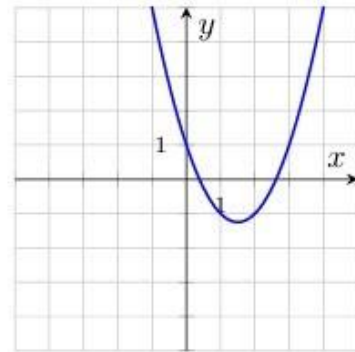
2. Quyida keltirilgan oltita funksiyadan uchtasining grafiklari rasmda tasvirlangan. Grafiklarga mos funksiyani toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = x^2 - 3x + 1$

3) $y = -x^2 - 3x + 1$

5) $y = -x + 2$

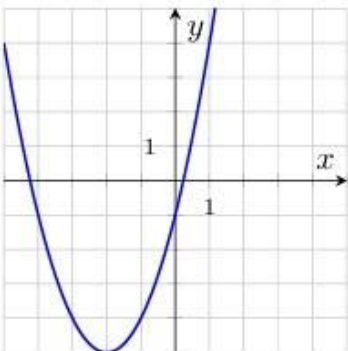
2) $y = \frac{3}{x}$

4) $y = x + 2$

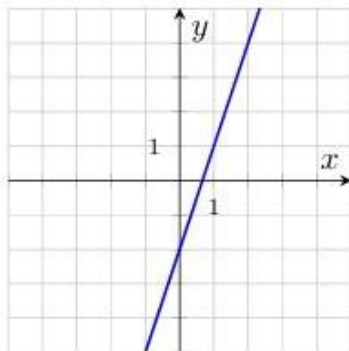
6) $y = -x + 2$

	A	B	C
Javob:	5	2	1

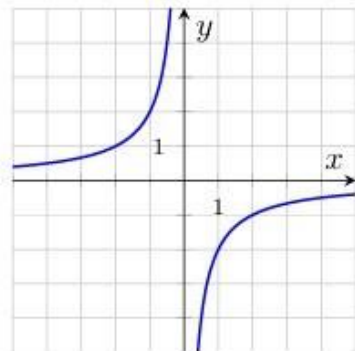
3. Quyida keltirilgan oltita funksiyadan uchtasining grafiklari rasmda tasvirlangan. Grafiklarga mos funksiyani toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = -\frac{2}{x}$

3) $y = 3x - 2$

5) $y = -3x - 2$

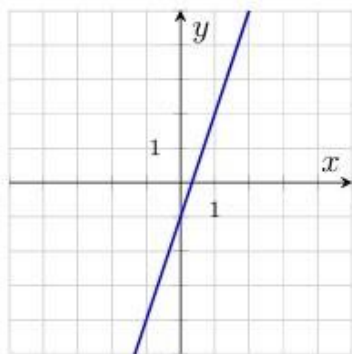
2) $y = -x^2 + 4x - 1$

4) $y = x^2 + 4x - 1$

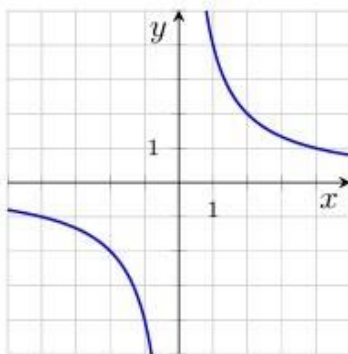
6) $y = \frac{2}{x}$

	A	B	C
Javob:	4	3	1

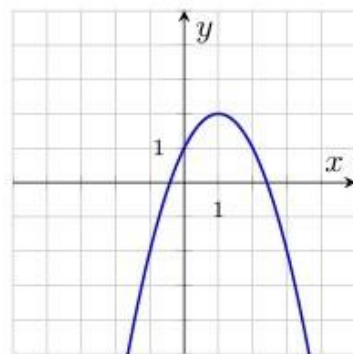
4. Quyida keltirilgan oltita funksiya uchrasining grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = \frac{4}{x}$

3) $y = x^2 + 2x + 1$

5) $y = -x^2 + 2x + 1$

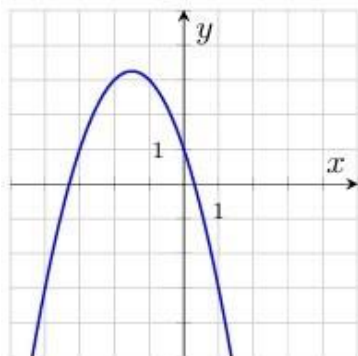
2) $y = 3x - 1$

4) $y = -3x - 1$

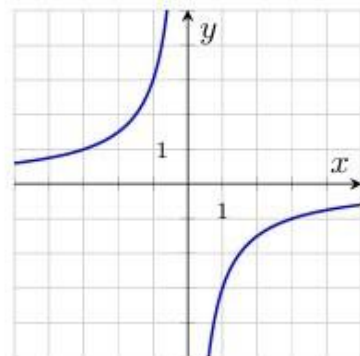
6) $y = \frac{4}{x}$

	A	B	C
Javob:	2	6	5

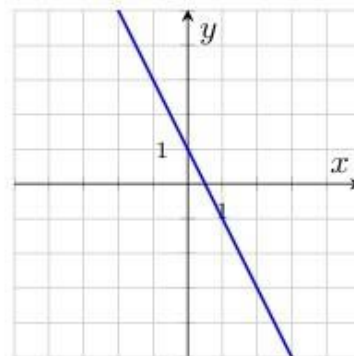
5. Quyida keltirilgan oltita funksiya uchrasining grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = -x^2 - 3x + 1$

3) $y = 2x + 1$

5) $y = -\frac{3}{x}$

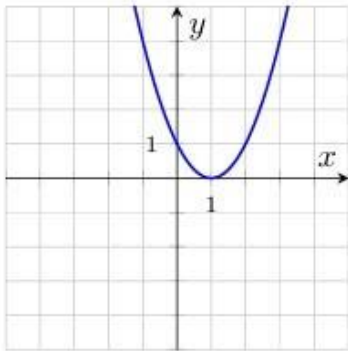
2) $y = \frac{3}{x}$

4) $y = -2x + 1$

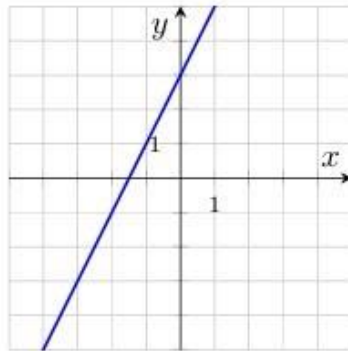
6) $y = x^2 - 3x + 1$

	A	B	C
Javob:	1	5	4

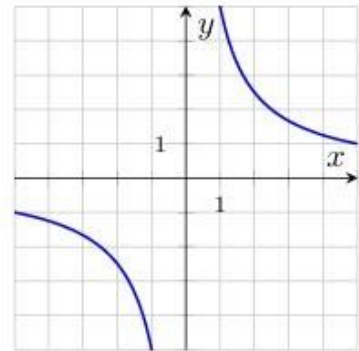
6. Quyida keltirilgan oltita funksiya uchta grafigi tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = -x^2 - 2x + 1$

3) $y = -2x + 3$

5) $y = x^2 - 2x + 1$

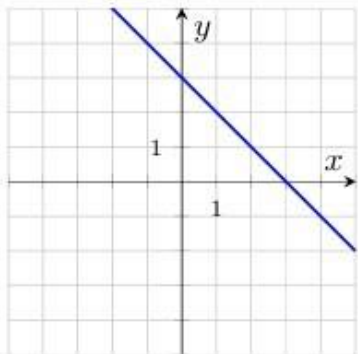
2) $y = 2x + 3$

4) $y = \frac{5}{x}$

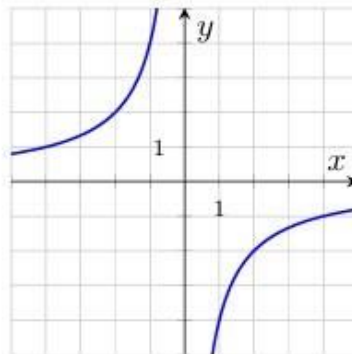
6) $y = -\frac{5}{x}$

	A	B	C
Javob:	5	2	4

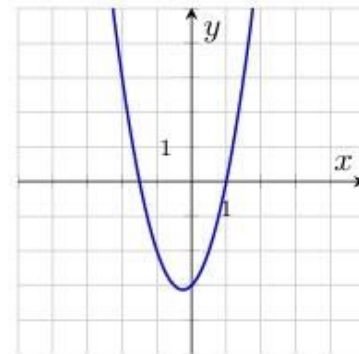
7. Quyida keltirilgan oltita funksiya uchta grafigi tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = -2x^2 + x - 3$

3) $y = -x + 3$

5) $y = -\frac{4}{x}$

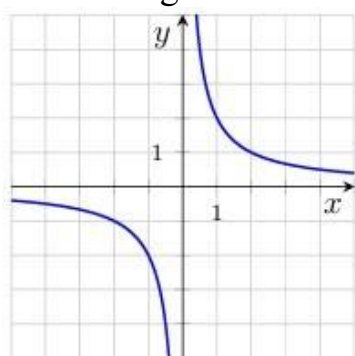
2) $y = \frac{4}{x}$

4) $y = 2x^2 + x - 3$

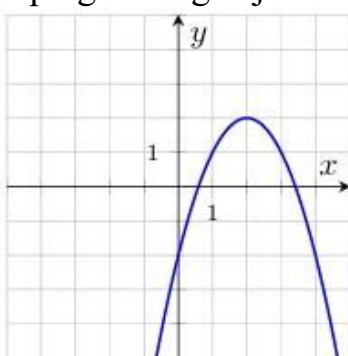
6) $y = x + 3$

	A	B	C
Javob:	3	5	4

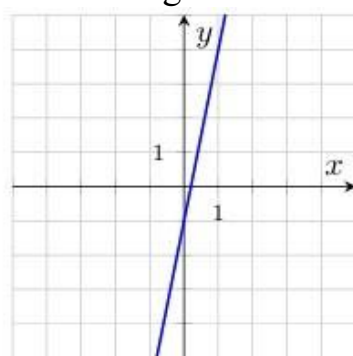
8. Quyida keltirilgan oltita funksiya uchta grafigi tasvirlangan. Grafiglarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = -\frac{2}{x}$

3) $y = -5x - 1$

5) $y = \frac{2}{x}$

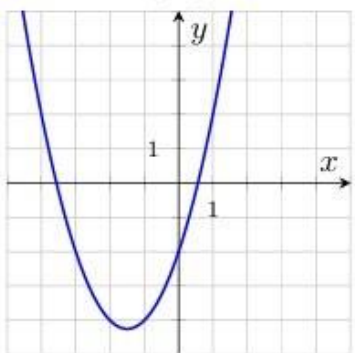
2) $y = 5x - 1$

4) $y = -x^2 + 4x - 2$

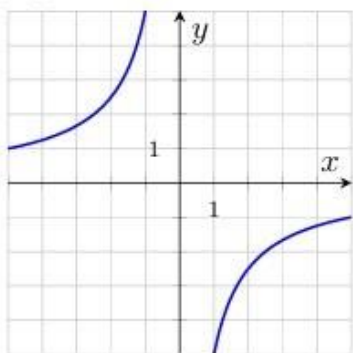
6) $y = x^2 + 4x - 2$

	A	B	C
Javob:	5	4	2

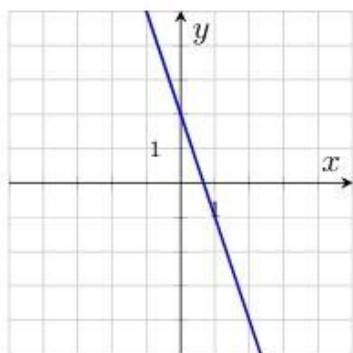
9. Quyida keltirilgan oltita funksiya uchta grafigi tasvirlangan. Grafiglarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = 3x + 2$

3) $y = -x^2 + 3x - 2$

5) $y = -\frac{5}{x}$

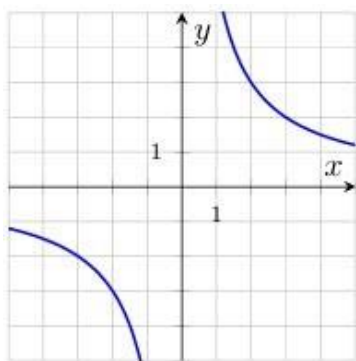
2) $y = \frac{5}{x}$

4) $y = -3x + 2$

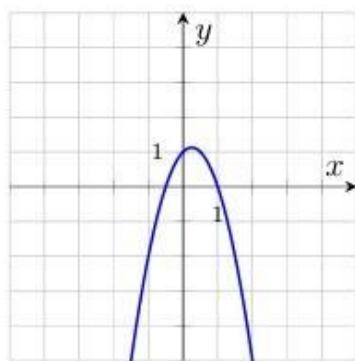
6) $y = x^2 + 3x - 2$

	A	B	C
Javob:	6	5	4

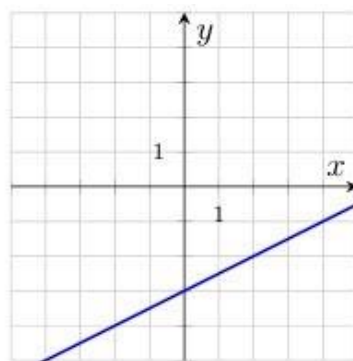
10. Quyida keltirilgan oltita funksiya uchrasining grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = \frac{6}{x}$

3) $y = -\frac{6}{x}$

5) $y = 0.5x - 3$

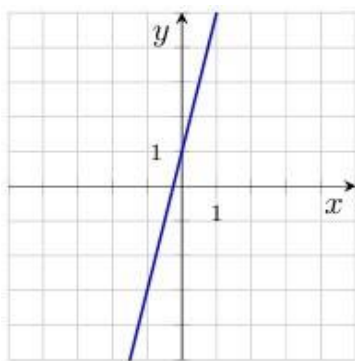
2) $y = 2x^2 + x + 1$

4) $y = 0.5x - 3$

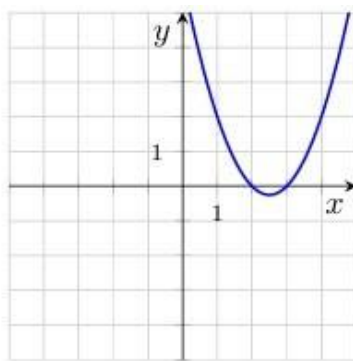
6) $y = -2x^2 + x + 1$

	A	B	C
Javob:	1	6	4

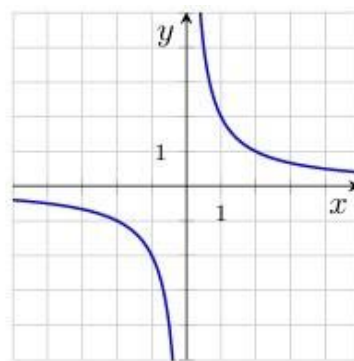
11. Quyida keltirilgan oltita funksiya uchrasining grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = 4x + 1$

3) $y = -\frac{2}{x}$

5) $y = -4x + 1$

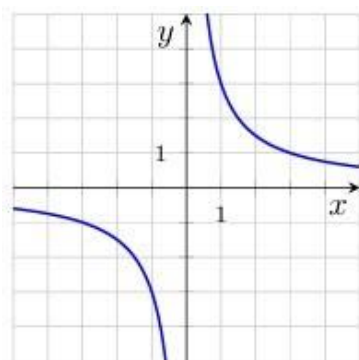
2) $y = \frac{2}{x}$

4) $y = -x^2 - 5x + 6$

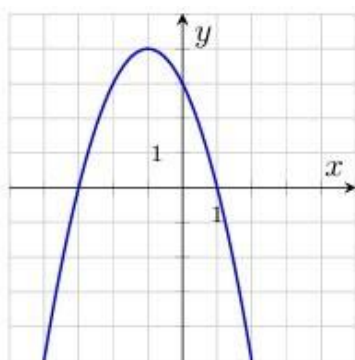
6) $y = -2x^2 + x + 1$

	A	B	C
Javob:	1	6	2

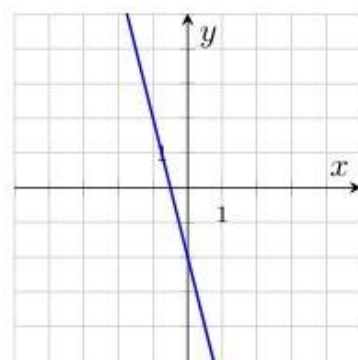
12. Quyida keltirilgan oltita funksiya uchta sinning grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = 4x - 2$

3) $y = -\frac{3}{x}$

5) $y = \frac{3}{x}$

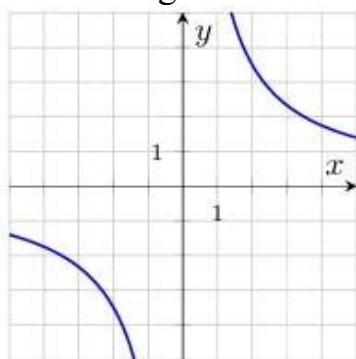
2) $y = -x^2 - 2x + 3$

4) $y = -4x - 2$

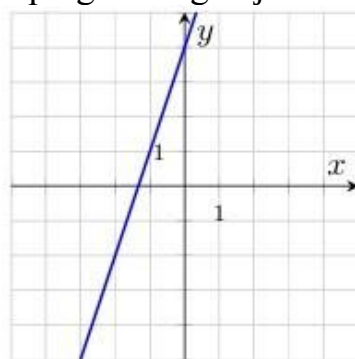
6) $y = x^2 - 2x + 3$

	A	B	C
Javob:	5	2	3

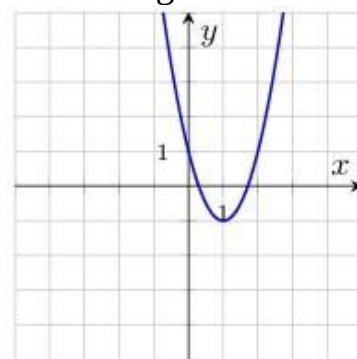
13. Quyida keltirilgan oltita funksiya uchta sinning grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = -2x^2 - 4x + 1$

3) $y = \frac{7}{x}$

5) $y = -\frac{7}{x}$

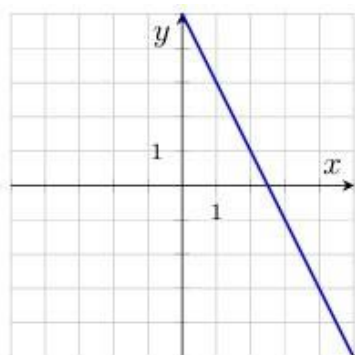
2) $y = 3x + 4$

4) $y = 2x^2 - 4x + 1$

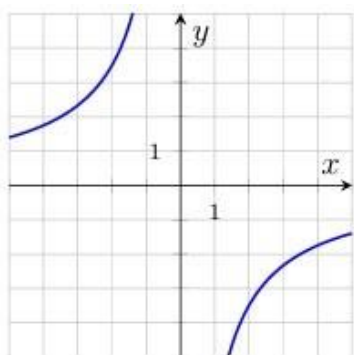
6) $y = -3x + 4$

	A	B	C
Javob:	3	2	4

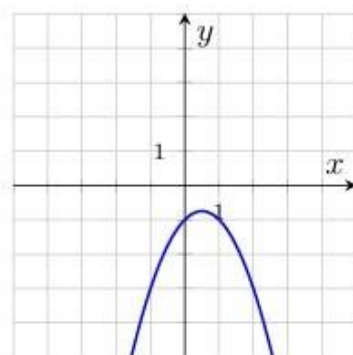
14. Quyida keltirilgan oltita funksiya uchta sinning grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = \frac{7}{x}$

3) $y = 2x + 5$

5) $y = -\frac{7}{x}$

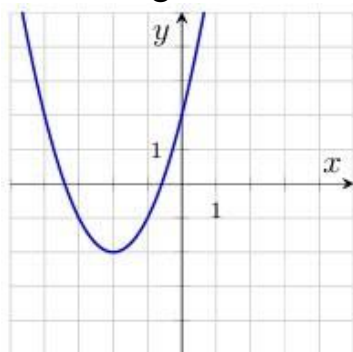
2) $y = -2x + 5$

4) $y = -x^2 + x - 1$

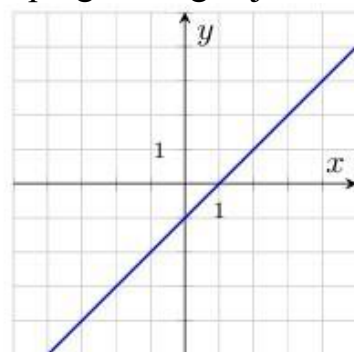
6) $y = x^2 + x - 1$

	A	B	C
Javob:	2	5	4

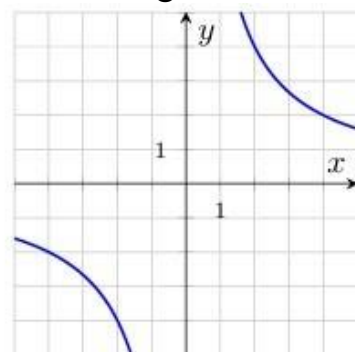
15. Quyida keltirilgan oltita funksiya uchta sinning grafiklari rasmda tasvirlangan. Grafiklarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = \frac{8}{x}$

3) $y = x^2 + 4x + 2$

5) $y = -x - 1$

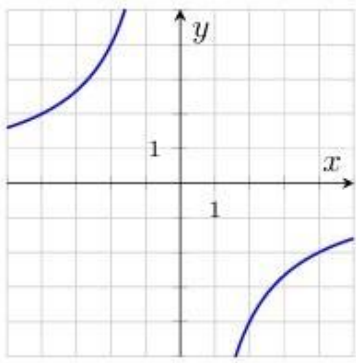
2) $y = x - 1$

4) $y = -\frac{8}{x}$

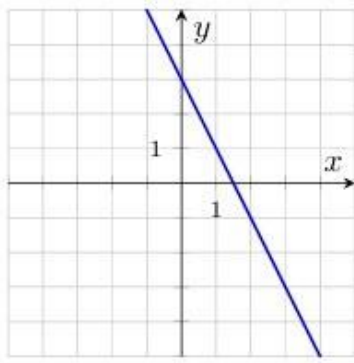
6) $y = -x^2 + 4x + 2$

	A	B	C
Javob:	3	2	1

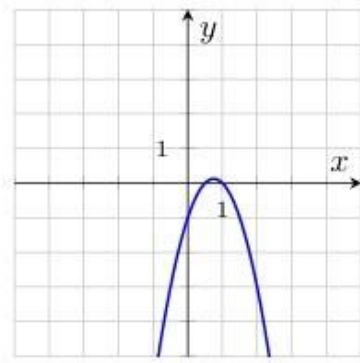
16. Quyida keltirilgan oltita funksiya uchta grafigi tasvirlangan. Grafiglarga mos funksiya toping va to'g'ri javoblarni mos keltiring.



A



B



C

Funksiyalar:

1) $y = 2x^2 + 3x - 1$

3) $y = \frac{8}{x}$

5) $y = -2x^2 + 3x - 1$

2) $y = 2x + 3$

4) $y = -2x + 3$

6) $y = -\frac{8}{x}$

	A	B	C
Javob:	6	4	5

V. Tenglamalar va ularning sistemalari

1. Tenglama ildizlaridan kattasini toping

$$x^2 + x = 12$$

J: $x_2 - 3$

2. Tenglamani yeching:

$$-(x + 6) = 2x + 18$$

J: $x = -8$

3. Tenglamalar sistemasini yeching va $x \cdot y$ ko'paytma qiymatini toping:

$$\begin{cases} x + y = 2 \\ 2x - y = -4 \end{cases}$$

J: $x = -\frac{2}{3} \quad y = 2\frac{2}{3}$

4. Tenglama ildizlari yig'indisini toping:

$$(x - 3)(3x + 15)(2x - 4) = 0$$

J: $3 + (-5) + 2 = 0$

5. Tenglama ildizlaridan kichigini toping:

$$x^2 = 2x + 24$$

J: $x = -4$

6. Tenglamani yeching:

$$4x + 8 = 2(x - 11)$$

J: $x = -15$

7. Tenglamalar sistemasini yeching va $x+y$ yig'indi qiymatini toping:

$$\begin{cases} x - y = 3 \\ 2x + y = 12 \end{cases}$$

J: $x = 5 \quad y = 2$

8. Tenglama ildizlari yig'indisini toping:

$$(x + 1)(2x + 16)(3x - 15) = 0$$

J: $-1 + (-8) + 5 = -4$

9. Tenglama ildizlaridan kattasini toping:

$$x^2 - 5x = 6$$

J: $x = 6$

10. Tenglamani yeching:

$$-(2x - 3) = x + 27$$

$$\text{J: } x = -8$$

11. Tenglamalar sistemasin yeching va $x \cdot y$ ko'paytma qiymatini toping:

$$\begin{cases} x + 2y = 7 \\ 2x - y = -1 \end{cases}$$

$$\text{J: } x = 1 \quad y = 3$$

12. Tenglama ildizlari yig'indisini toping:

$$(x - 7)(3x + 12)(2x - 10) = 0$$

$$\text{J: } 7 + (-4) + 5 = 8$$

13. Tenglama ildizlaridan kichigini toping:

$$x^2 = -3x + 10$$

$$\text{J: } x = -5$$

14. Tenglamani yeching:

$$4x + 8 = 2(x - 7)$$

$$\text{J: } x = -11$$

15. Tenglamalar sistemasini yeching va $x+y$ yig'indi qiymatini toping:

$$\begin{cases} x - y = -1 \\ x + 2y = 5 \end{cases}$$

$$\text{J: } x = 1 \quad y = 2$$

16. Tenglama ildizlari yig'indisini toping:

$$(x + 2)(4x + 32)(2x - 12) = 0$$

$$\text{J: } -2 + (-8) + 6 = -4$$

VI. Tengsizliklar va ularning sistemalari

1. Tengsizliklar sistemasini yeching:

$$\begin{cases} 2x - 2 \geq x - 8 \\ -2x + 7 < -5 \end{cases}$$

$$\text{J: } x > 6$$

2. Tengsizlikni yeching:

$$(2x - 6)(x + 1) < 0$$

$$\text{J: } -1 < x < 3$$

3. Tengsizlikni yeching:

$$-1 \leq -2x + 3 < 5$$

$$\text{J: } -1 < x \leq 2$$

4. Tengsizlikni yeching:

$$x^2 + 10x \geq 0$$

$$\text{J: } x \in (-\infty; -10] \cup [0; +\infty)$$

5. Tengsizliklar sistemasini yeching:

$$\begin{cases} 3x - 5 \geq x - 15 \\ -2x + 1 > -1 \end{cases}$$

$$\text{J: } 1 < x$$

6. Tengsizlikni yeching:

$$(x - 3)(4x + 8) \geq 0$$

$$\text{J: } x \in (-\infty; -2] \cup [3; +\infty)$$

7. Tengsizlikni yeching:

$$-2 \leq -3x + 4 < 16$$

$$\text{J: } -4 < x \leq 2$$

8. Tengsizlikni yeching:

$$x^2 - 25 < 0$$

$$\text{J: } x \in (-5; 5)$$

9. Tengsizliklar sistemasini yeching:

$$\begin{cases} 2x - 4 \geq x - 10 \\ -4x + 1 < -7 \end{cases}$$

$$\text{J: } x > 2$$

10. Tengsizlikni yeching:

$$(5x - 15)(x + 4) < 0$$

$$\text{J: } x \in (-4; 3)$$

11. Tengsizlikni yeching:

$$-3 \leq -2x + 7 < 13$$

$$\text{J: } -3 < x \leq 5$$

12. Tengsizlikni yeching:

$$x^2 - 4x \leq 0$$

$$\text{J: } x \in [0; 4]$$

13. Tengsizliklar sistemasini yeching:

$$\begin{cases} 3x - 3 \geq x - 11 \\ -2x + 5 > -7 \end{cases}$$

$$\text{J: } -4 \leq x < 5$$

14. Tengsizlikni yeching:

$$(x - 6)(3x + 12) \geq 0$$

$$\text{J: } x \in (-\infty; -4] \cup [6; +\infty)$$

15. Tengsizlikni yeching:

$$-5 \leq -3x + 1 < 16$$

$$\text{J: } -5 < x \leq 2$$

16. Tengsizlikni yeching:

$$x^2 - 49 > 0$$

$$\text{J: } x \in (-\infty; -7) \cup (7; +\infty)$$

VII. Kvadrat funksiyalar

1. $y = -2x^2 + 8x - 6$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

1. Funksiya grafigi koordinata tekisligining qaysi choraklaridan o'tishini;
2. Funksiya grafigining koordinata o'qlari bilan kesishish nuqtalari;
3. Funksiyaning aniqlanish sohasi;
4. Funksiyaning qiymatlar to'plami;
5. Funksiyaning o'sish va kamayish oraliqlari.

$$\text{J: } 1) \text{ I, III, IV } 2) x_1 = 1, x_2 = 3, y = -6 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 2) \\ 5) \uparrow y \in (-\infty; 2), \downarrow y \in (2; +\infty)$$

2. $y = x^2 + 2x - 8$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$\text{J: } 1) \text{ I, II, III, IV } 2) x_1 = -4, x_2 = 2, y = -8 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-9; +\infty) \\ 5) \downarrow y \in (-\infty; -1), \uparrow y \in (-1; +\infty)$$

3. $y = 2x^2 - 4x - 6$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$\text{J: } 1) \text{ I, II, III, IV } 2) x_1 = -1, x_2 = 3, y = -6 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-8; +\infty) \\ 5) \downarrow y \in (-\infty; 1), \uparrow y \in (1; +\infty)$$

4. $y = -x^2 + 4x + 5$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$\text{J: } 1) \text{ I, II, III, IV } 2) x_1 = -1, x_2 = 5, y = 5 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 9) \\ 5) \uparrow y \in (-\infty; 2), \downarrow y \in (2; +\infty)$$

5. $y = x^2 + 4x + 3$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$\text{J: } 1) \text{ I, II, III } 2) x_1 = -3, x_2 = -1, y = 3 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-1; +\infty)$$

$$5) \downarrow y \in (-\infty; -2), \uparrow y \in (-2; +\infty)$$

6. $y = -x^2 - 2x + 8$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, III, IV} \quad 2) x_1 = -4, x_2 = 2, y = 8 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 9) \\ 5) \uparrow y \in (-\infty; -1), \downarrow y \in (-1; +\infty)$$

7. $y = x^2 - 6x + 8$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, IV} \quad 2) x_1 = 2, x_2 = 4, y = 8 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-1, +\infty) \\ 5) \downarrow y \in (-\infty; 3), \uparrow y \in (3; +\infty)$$

8. $y = 2x^2 + 8x + 6$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, III} \quad 2) x_1 = -3, x_2 = -1, y = 6 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-1, +\infty) \\ 5) \downarrow y \in (-\infty; -2), \uparrow y \in (-2; +\infty)$$

9. $y = -x^2 + 6x - 5$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, III, IV} \quad 2) x_1 = 1, x_2 = 5, y = -5 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 4) \\ 5) \uparrow y \in (-\infty; 3), \downarrow y \in (3; +\infty)$$

10. $y = -x^2 - 6x - 8$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ II, III, IV} \quad 2) x_1 = -4, x_2 = -1, y = 3 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 1) \\ 5) \uparrow y \in (-\infty; -3), \downarrow y \in (-3; +\infty)$$

11. $y = 2x^2 + 4x - 6$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, III, IV} \quad 2) x_1 = -3, x_2 = 1, y = -6 \quad 3) D \in (-\infty; +\infty) \quad 4) (-8, +\infty) \\ 5) \downarrow y \in (-\infty; -1), \uparrow y \in (-1; +\infty)$$

12. $y = x^2 + 4x - 5$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, III, IV} \quad 2) x_1 = -5, x_2 = 1, y = -5 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-9, +\infty) \\ 5) \downarrow y \in (-\infty; -2), \uparrow y \in (-2; +\infty)$$

13. $y = -x^2 + 2x + 8$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, III, IV} \quad 2) x_1 = -2, x_2 = 4, y = 8 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 9) \\ 5) \uparrow y \in (-\infty; 1), \downarrow y \in (1; +\infty)$$

14. $y = -x^2 + 2x + 3$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

$$J: \quad 1) \text{ I, II, III, IV} \quad 2) x_1 = -1, x_2 = 3, y = 3 \quad 3) D \in (-\infty; +\infty) \quad 4) x \in (-\infty; 4) \\ 5) \uparrow y \in (-\infty; 1), \downarrow y \in (1; +\infty)$$

15. $y = x^2 + 6x + 5$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

- J: 1) I, II, III 2) $x_1 = -5, x_2 = -1, y = 5$ 3) $D \in (-\infty; +\infty)$ 4) $(-4, +\infty)$
5) $\downarrow y \in (-\infty; -3), \uparrow y \in (-3; +\infty)$

16. $y = x^2 + 2x - 3$ funksiyaning grafigini chizing. Grafikdan foydalanib, quyidagilarni aniqlang:

- J: 1) I, II, III, IV 2) $x_1 = 1, x_2 = -3, y = -3$ 3) $D \in (-\infty; +\infty)$ 4) $(-1, +\infty)$
5) $\downarrow y \in (-\infty;), \uparrow y \in (; +\infty)$

VIII. Trigonometrik ayniyatlar

1. Agar $\cos \alpha = -\frac{\sqrt{21}}{5}$ va $\frac{\pi}{2} < \alpha < \pi$ bo'lsa, $\sin \alpha$ ning qiymatini toping. J: $\sin \alpha = \frac{2}{5}$
2. Ifodaning qiymatini toping: $\operatorname{tg} 10^\circ \cdot \operatorname{ctg} 10^\circ + 4 \sin 30^\circ$. J: 3
3. Agar $\sin \alpha = -\frac{\sqrt{15}}{4}$ va $-\frac{\pi}{2} < \alpha < 0$ bo'lsa, $\cos \alpha$ ning qiymatini toping. J: $\cos \alpha = \frac{1}{4}$
4. Ifodaning qiymatini toping: $\sqrt{3} \operatorname{tg} 60^\circ - \sin^2 20^\circ - \cos^2 20^\circ$. J: 2
5. Agar $\cos \alpha = -\frac{3}{5}$ va $\pi < \alpha < \frac{3\pi}{2}$ bo'lsa, $\sin \alpha$ ning qiymatini toping. J: $\sin \alpha = -\frac{4}{5}$
6. Ifodaning qiymatini toping: $\operatorname{tg} 80^\circ \cdot \frac{\cos 80^\circ}{\sin 80^\circ} + 6 \cos 120^\circ$ J: -2
7. Agar $\sin \alpha = \frac{\sqrt{51}}{10}$ va $\frac{\pi}{2} < \alpha < \pi$ bo'lsa, $\cos \alpha$ ning qiymatini toping. J: $\cos \alpha = -\frac{7}{10}$
8. Ifodaning qiymatini toping: $\sin^2 110^\circ + \cos^2 110^\circ + 5\sqrt{2} \sin 45^\circ$. J: 6
9. Agar $\cos \alpha = -0,6$ va $\frac{\pi}{2} < \alpha < \pi$ bo'lsa, $\sin \alpha$ ning qiymatini toping. J: $\sin \alpha = 0,8$
10. Ifodaning qiymatini toping: $\operatorname{ctg} 80^\circ \cdot \operatorname{tg} 80^\circ + 12 \cos 60^\circ$. J: 7
11. Agar $\sin \alpha = -\frac{\sqrt{7}}{4}$ va $-\frac{\pi}{2} < \alpha < 0$ bo'lsa, $\cos \alpha$ ning qiymatini toping. J: $\cos \alpha = \frac{3}{4}$
12. Ifodaning qiymatini toping: $2\sqrt{3} \operatorname{ctg} 30^\circ - \sin^2 140^\circ - \cos^2 140^\circ$. J: 5
13. Agar $\cos \alpha = -\frac{2\sqrt{6}}{5}$ va $\pi < \alpha < \frac{3\pi}{2}$ bo'lsa, $\sin \alpha$ ning qiymatini toping. J: $\sin \alpha = -\frac{1}{5}$
14. Ifodaning qiymatini toping: $\operatorname{ctg} 40^\circ \cdot \frac{\sin 40^\circ}{\cos 40^\circ} - 2 \cos 120^\circ$ J: 2
15. Agar $\sin \alpha = \frac{\sqrt{91}}{10}$ va $\frac{\pi}{2} < \alpha < \pi$ bo'lsa, $\cos \alpha$ ning qiymatini toping. J: $\cos \alpha = -\frac{3}{10}$
16. Ifodaning qiymatini toping: $\sin^2 200^\circ + \cos^2 200^\circ + 3\sqrt{2} \cos 45^\circ$ J: 4

IX. Amaliy masalalar yechishda algebraik

1. Avtoparkda jami 25 ta avtomobil (yengil va yuk avtomobillari) bor. Yengil avtomobil bakiining sig'imi 50 litr, yuk avtomobiliniki esa 120 litr. Avtoparkdagi barcha avtomobillarni yoqilg'i bilan to'ldirish uchun jami 2090 litr yoqilg'i sarflandi. Avtoparkda nechta yengil va nechta yuk avtomobili bor?

- J: yengil-13, yuk-12

2. Birinchi tizim administratori bir soatda ikkinchi administratorga qaraganda 3 ta kompyuterga ko'proq xizmat ko'rsatadi. Ular birgalikda 5 soatda 55 ta kompyuterga xizmat ko'rsatishdi. Har bir administrator bir soatda nechta kompyuterga xizmat ko'rsatadi?

J: birinchi-7 ta, ikkinchi-4 ta

3. Portda yuk ko'tarish quvvati turlicha bo'lgan ikkita kran konteynerlarni tushirmoqda. Birinchi kran bir soatda ikkinchisiga qaraganda 5 ta konteynerga ko'proq tushiradi. Ular 6 soat davomida birgalikda ishlab, jami 270 ta konteynerni tushirishdi. Har bir kran bir soatda nechta konteyner tushiradi?

J: birinchi-25 ta, ikkinchi-20 ta

4. Ikki guruh optik tolali kabel yotqizmoqda. Birinchi guruh bir kunda ikkinchisiga qaraganda 100 m kam kabel yotqizadi. 5 kun davomida birgalikda ishlab, jami 4000 m kabel yotqizishdi. Har bir guruhning kunlik ish unumdorligini toping.

J: birinchi-350, ikkinchi-450

5. Laboratoriyada ikki qurilma ishladi. Birinchi qurilma 5 soat, ikkinchisi 3 soat ishlaganda jami 13 kVt·soat elektr energiyasi sarflandi. Boshqa kuni birinchi qurilma 3 soat, ikkinchisi 6 soat ishlaganda jami 12 kVt·soat energiya sarflandi. Har bir qurilmaning quvvatini toping.

J: birinchi-2 kVt·soat, ikkinchi-1 kVt·soat

6. Avtomatlashtirilgan stanokda A va B turdagi detallar ishlab chiqariladi. Bitta A detalni tayyorlash uchun 2 daqiqa, B detal uchun 3 daqiqa vaqt ketadi. 8 soatlik smenada stanok jami 180 ta detal ishlab chiqardi. Har bir turdan nechta detal tayyorlangan?

J: A detal 120 ta, B detal 60 ta.

7. Zavodda ikkita yig'ish liniyasi bir xil mahsulot ishlab chiqaradi. Ikkinchi liniya bir soatda birinchisiga qaraganda 3 ta mahsulotga ko'proq yig'adi. 5 soatlik birgalikdagi ish davomida ular 155 ta mahsulot yig'ishdi. Har bir liniya bir soatda nechta mahsulot yig'adi?

J: birinchi liniya-14 ta, ikkinchi liniya-17 ta

8. Taksoparkda jami 30 ta elektromobil bor: yengil avtomobillar va mikroavtobuslar. Yengil avtomobilni zaryadlash uchun 50 kVt·soat, mikroavtobusni zaryadlash uchun esa 120 kVt·soat energiya talab etiladi. Butun parkdagi barcha elektromobillarni zaryadlash uchun 1920 kVt·soat energiya sarflandi. Parkda nechta yengil avtomobil va nechta mikroavtobus bor?

J: yengil avto-24 ta, mikroavtobus-6 ta.

9. Ikki guruh avtomobil yo'lini ta'mirlamoqda. Birinchi guruh bir kunda ikkinchisiga qaraganda 50 m kam yo'lni ta'mirlaydi. 4 kunlik birgalikdagi ish davomida ular jami 1800 m yo'lni ta'mirlashdi. Har bir guruhning bir kunlik ish unumdorligini toping.

J: birinchi guruh-200 m, ikkinchi guruh-250 m

10. Birinchi apparat 5 soat, ikkinchi apparat 3 soat ishlaganda jami 270 litr bo'yoq sarflandi. Boshqa kuni birinchi apparat 3 soat, ikkinchisi 6 soat ishlaganda 330 litr bo'yoq sarflandi. Har bir apparat bir soatda nechta litr bo'yoq sarflaydi?

J: birinchi apparat-30, ikkinchi apparat-40 ta.

11. Birinchi stanok 4 soat, ikkinchi stanok 6 soat ishlaganda jami 32 ta detal tayyorlandi. Keyingi kuni birinchi stanok 6 soat, ikkinchisi 2 soat ishlaganda jami 34 ta detal tayyorlandi. Har bir stanok bir soatda nechta detal ishlab chiqaradi?

J: birinchi stanok-5, ikkinchi stanok-2 ta detal.

12. Ikki xil zaryadlovchi qurilma ishlatiladi. Agar birinchi qurilma 2 soat, ikkinchisi 5 soat ishlaganda jami 23 kVt·soat energiya sarflanadi. Agar birinchi qurilma 4 soat, ikkinchisi 3 soat ishlaganda 25 kVt·soat energiya sarflanadi. Har bir zaryadlovchi qurilmaning quvvatini toping.

J: birinchi qurilma-4 kVt·soat, ikkinchi qurilma-3 kVt·soat

13. Ikki server ma'lumotlarni qayta ishlamoqda. Birinchi server 7 soat, ikkinchisi 4 soat ishlaganda jami 125 GB ma'lumot qayta ishlandi. Boshqa safar birinchi server 5 soat, ikkinchisi 8 soat ishlagandi 115 GB ma'lumot qayta ishlandi. Har bir server bir soatda necha gigabayt ma'lumot qayta ishlaydi?

J: Birinchi server-15 gb, ikkinchi server-5gb

14. Ikki nasos yordamida suv tortilmoqda. Birinchi nasos 6 daqiqa, ikkinchi nasos 2 daqiqa ishladi va 1600 l suv tortib olindi. Boshqa safar birinchi nasos 3 daqiqa, ikkinchisi 5 daqiqa ishladi va 1300 l suv tortib olindi. Har bir nasos bir daqiqada necha litr suv tortadi?

J: birinchi nasos-225 l, ikkinchi nasos-125 litr

15. Mahalliy tarmoq yotqizish uchun 50 ta kabel to'plami sotib olindi. Bu to'plamlarning bazilarida 100 m, boshqalarida 150 m kabel bor. Barcha kabellarning umumiy uzunligi 6000 m ni tashkil etdi. Har bir turdagi ya'ni 100 m li va 150 m li to'plamlardan nechta sotib olingan?

J: 100 m likdan 30 ta, 150 m likdan 20 ta.

16. Kompyuter sinfida jami 40 ta kompyuter bor: ularning bir qismida A darajali, boshqalarida B darajali videokartalar o'rnatilgan. A darajali videokarta 75 Vt, B darajali videokarta esa 120 Vt quvvat iste'mol qiladi. Barcha kompyuterlar bir vaqtda ishlaganda umumiy energiya iste'moli 3720 Vt ga teng. Har bir turdagi videokartaga ega kompyuterlar sonini toping.

J: A darjadagi -24 ta, B darjadagi 16 ta

X. Progressiyalar

1. Arifmetik progressiyaning uchinchi hadi 6 ga, beshinchi hadi esa 12 ga teng. Dastlabki oltita hadining yig'indisini toping.

J: $S_6 = 30$

2. Geometrik progressiyaning uchinchi hadi 60 ga, to'rtinchi hadi esa 30 ga teng. Dastlabki beshta hadining yig'indisini toping.

J: $S_5 = 465$

3. Arifmetik progressiyaning beshinchi hadi 12 ga, progressiya ayirmasi 4 ga teng. Dastlabki yettita hadining yig'indisini toping.

J: $S_7 = 56$

4. Geometrik progressiyaning ikkinchi hadi 4 ga, progressiya maxraji 2 ga teng. Dastlabki beshta hadining yig'indisini toping.

$$J: S_5 = 62$$

5. Arifmetik progressiyaning birinchi hadi 20 ga teng, beshinchi va uchinchi hadlar ayirmasi 6 ga teng. Progressiyaning o'ninchi hadini toping.

$$J: a_5 = 47$$

6. Geometrik progressiyaning birinchi hadi 10 ga teng, beshinchi hadi ikkinchi haddan 27 marta katta. Progressiyaning beshinchi hadini toping.

$$J: b_5 = 810$$

7. Arifmetik progressiyaning ikkinchi hadi 20 ga, to'rtinchi hadi esa 16 ga teng. Dastlabki beshta hadining yig'indisini toping.

$$J: S_5 = 90$$

8. Geometrik progressiyaning ikkinchi hadi 6 ga, uchinchi hadi esa 18 ga teng. Dastlabki to'rtta hadining yig'indisini toping.

$$J: S_4 = 80$$

9. Arifmetik progressiyaning sakkizinchi hadi 40 ga, progressiya ayirmasi 5 ga teng. Dastlabki yettita hadining yig'indisini toping.

$$J: S_7 = 140$$

10. Geometrik progressiyaning ikkinchi hadi 10 ga, progressiya maxraji 5 ga teng. Dastlabki to'rtta hadining yig'indisini toping.

$$J: S_4 = 312$$

11. Arifmetik progressiyaning birinchi hadi 8 ga teng, oltinchi va ikkinchi hadlar ayirmasi 16 ga teng. Progressiyaning beshinchi hadini toping.

$$J: a_5 = 24$$

12. Geometrik progressiyaning birinchi hadi 7 ga teng, oltinchi hadi uchinchi haddan 8 marta katta. Progressiyaning to'rtinchi hadini toping.

$$J: b_4 = 56$$

13. Arifmetik progressiyaning uchinchi hadi 6 ga, beshinchi hadi esa 12 ga teng. Dastlabki o'nta hadining yig'indisini toping.

$$J: S_{10} = 225$$

14. Geometrik progressiyaning ikkinchi hadi 40 ga, uchinchi hadi esa 20 ga teng. Dastlabki to'rtta hadining yig'indisini toping.

$$J: S_4 = 150$$

15. Arifmetik progressiyaning birinchi hadi 5 ga teng, beshinchi va uchinchi hadlar ayirmasi 40 ga teng. Progressiyaning oltinchi hadini toping.

$$J: a_6 = 105$$

16. Geometrik progressiyaning birinchi hadi 20 ga teng, beshinchi hadi ikkinchi haddan 8 marta katta. Progressiyaning to'rtinchi hadini toping.

$$J: b_4 = 160$$

XI. Kombinatorika elementlari

1. Do'konda 5 xil chizg'ich va 4 xil o'chirg'ich sotiladi. Bitta chizg'ich va bitta o'chirg'ichdan iborat to'plamni necha xil usulda sotib olish mumkin?

J: 20

2. Do'konda 4 xil rangdagi sharlar sotiladi: qizil, ko'k, sariq va yashil. Ranglari har xil bo'lgan ikki sharni necha xil usulda sotib olish mumkin?

J: 6

3. Tushgacha kafeda ichimlik va bitta non mahsulotidan (kulcha, qatlama non, patir) iborat «Yengil nonushta» to'plamini buyurtma qilish mumkin. Kafeda 7 xil ichimlik mavjud. «Yengil nonushta»ning nechta turli varianti mavjud?

J: 21

4. Yangi avtomobil xarid qilinayotganda xaridor avtomobil rangini 5 ta taklif etilgan variantdan birini tanlashi mumkin. Shuningdek, g'ildirak turi sifatida po'lat yoki alyuminiy g'ildiraklardan biri tanlanadi. Avtomobilning «rang + g'ildirak turi» tanlovlari asosida nechta turli variant tuzish mumkin?

J: 10

5. Menyuda pitsaning taklif etilgan 5 ta turidan (zaytun, qo'ziqorin, qalampir, qiyma go'sht, nok) ikkitasini buyurtma qilish mumkin. Necha xil buyurtma qilish mumkin?

J: 10

6. Sayohatga 6 nafar kishi chiqdi. Kechki ovqatni tayyorlash uchun ulardan ikki nafar kishi mas'ul etib belgilanadi. Ikki kishini necha xil usulda tanlash mumkin?

J: 15

7. Oshxonada biznes-lanch birinchi taomdan (osh, lag'mon, norin) va ikkinchi taomdan (garnirli go'sht, baliq, tovuq) iborat. Necha turli biznes-lanch variantini tuzish mumkin?

J: 9

8. Chamadon qulf kodi ikki raqamdan iborat. Birinchi raqam — toq, ikkinchi raqam — juft. Shunday kodlarning nechta turli varianti mavjud?

J: 25

9. Robot yig'ish uchun bitta plata va bitta korpus tanlanadi. To'plamda 6 xil plata va 3 xil korpus (kichik, o'rta, katta) mavjud. Har bir qismdan bittadan tanlab, nechta turli robot yig'ish mumkin?

J: 18

10. Maktab shaxmat turniri finalida 4 nafar shaxmatchi ishtirok etdi. Har bir shaxmatchi boshqalarning har biri bilan 4 tadan partiya o'ynadi (2 marta oq donalarda va 2 marta qora donalarda). Jami nechta partiya o'ynalgan?

J: 6

11. Do'konda 8 xil rangdagi va 3 xil o'lchamdagi (S, M, L) futbolkalar sotiladi. Har bir rangda barcha o'lchamlar mavjud. Do'konda jami nechta turli futbolka mavjud?

J: 24

12. Yo'lovchi taksining uch xonali raqami 7 raqami bilan boshlanishini eslab qoldi.

Raqamning qolgan ikki xonasi juft raqamlardan iborat bo'lib, ular o'zaro turli. Shunday uch xonali raqamlarning nechta turli varianti mavjud?

J: 20

13. Menyuda «go'sht va pishloq»li pitsani buyurtma qilish mumkin. Tanlash uchun 7 xil pishloq va 2 xil go'sht turi mavjud. Nechta turli « go'sht va pishloq» pitsa buyurtma qilish mumkin?

J: 14

14. Turnirda 5 ta voleybol jamoasi ishtirok etdi. Har bir jamoa boshqa har bir jamoa bilan bittadan o'yin o'tkazdi. Jami nechta o'yin o'ynalgan?

J: 10

15. Tushgacha kafeda ichimlik va bitta somsadan (tovuqli, go'shtli, kartoshkali yoki qovoqli) iborat «Xayrli tong!» to'plamini buyurtma qilish mumkin. Kafeda 4 xil ichimlik mavjud. «Xayrli tong!» to'plamining nechta turli varianti mavjud?

J: 16

16. Kirish uchun mo'ljallangan kalit-karta kodi ikki belgidan iborat: birinchi belgi — harf (K, L, M), ikkinchi belgi — raqam (1, 2, 3, 4). Nechta turli kalit-karta kodi mavjud?

J: 12

XII. Ehtimollik va statistika

1. Taksi firmasida hozirda 20 ta oq, 6 ta qora va 4 ta kulrang avtomobil bo'sh turibdi. Chaqiruv bo'yicha buyurtmachiga eng yaqin bo'lgan avtomobillardan biri tasodifiy ravishda yo'lga chiqdi. Buyurtmachiga qora taksi kelish ehtimolini toping.

J: 20%

2. Feruz otasi bilan charxpalak attraksionida aylanishga qaror qildi. Charxpalakda jami 20 ta kabina bo'lib, ulardan 3 tasi ko'k, 10 tasi yashil, qolganlari qizil rangda. Kabinalar navbat bilan platformaga keladi. Feruzning qizil kabinada aylanish ehtimolini toping.

J: 35%

3. Imtihonda 50 ta bilet bor. Aziz ulardan 9 tasini o'rganmagan. Azizga o'rganilgan bilet ushish ehtimolini toping.

J: 82%

4. Ota-onalar qo'mitasi yil yakuni munosabati bilan bolalarga sovg'a sifatida pazllar xarid qildi: 8 tasi avtomobillar tasviri bilan, 10 tasi shahar manzaralari bilan va 6 tasi multfilm qahramonlari bilan. Sovg'alar tasodifiy taqsimlanadi. Malikaga multfilm qahramoni tasvirlangan pazl tushish ehtimolini toping.

J: 25%

5. O'yin boshlanishidan oldin hakam qaysi jamoa to'pni birinchi bo'lib o'yinga kiritishini aniqlash uchun tanga tashlaydi. «Paxtakor» jamoasi ketma-ket ikki o'yin o'tkazadi: avval «Nasaf», so'ng «Bunyodkor» jamoasi bilan. Har ikkala o'yinda ham birinchi bo'lib to'pni

o'yinga kiritish huquqi «Paxtakor»ga berilish ehtimolini toping.

J: 25%

6. Chempionatda AQSh, Xitoy, O'zbekiston, Fransiya va Germaniya jamoalari ishtirok etmoqda. Jamoa vakillariga raqamlar tasodifiy tarzda beriladi. 5-raqam Osiyo qit'asiga mansub jamoaga to'g'ri kelish ehtimolini toping.

J: 40%

7. Sinf rahbari o'quvchilarni taqdirlash uchun 20 ta ruchka sotib oldi: 8 tasi «Ikat» naqshi bilan, 10 tasi Toshkent manzarasi bilan va 2 tasi Navro'z ramzi bilan bezatilgan. Sovg'alar tasodifiy taqsimlanadi. Shahzodga «Ikat» naqshli ruchka tushish ehtimolini toping.

J: 40%

8. Toshkentdagi universitet o'qituvchisi guruhda navbatchini aniqlash uchun tasodifiy sonlar generatoridan foydalanadi. Guruhda 25 nafar talaba bo'lib, ulardan 10 nafari yigitlar. Bugun qiz talaba navbatchi bo'lish ehtimolini toping.

J: 60%

9. Toshkent universitetida abituriyentlar test topshirish uchun tasodifiy ravishda uchta auditoriyaga taqsimlanadi. Avval ikki auditoriyaga 60 nafardan kishi joylashtiriladi, qolganlari zaxira auditoriyaga o'tkaziladi. Birinchi kuni testga 150 nafar abituriyent keldi.

Bahodirning zaxira auditoriyaga tushish ehtimolini toping.

J: 20%

10. «Samarqand» avtovokzalida bir nechta avtobuslar jo'nab ketishga tayyor: 2 tasi Toshkentga, 3 tasi Buxoroga, 5 tasi Shahrisabzga. Dispetcher navbatdan tashqari tekshiruv uchun avtobusni tasodifiy tanlaydi. Tanlangan avtobusning Buxoro yoki Shahrisabz yo'nalishida bo'lish ehtimolini toping.

J: 80%

11. Toshkent genetik markazi laboratoriyasida 12 nafar olim va 8 nafar laborant ishlaydi. Ular orasida tasodifiy tarzda dam olish maskaniga 4 ta kasaba uyushma yo'llanmasi o'ynaladi. Samiraning yo'llanmalardan birini qo'lga kiritish ehtimolini toping.

J: 20%

12. Ifodali o'qish tanlovida 10 nafar o'quvchi ishtirok etadi. Qur'a tashlash orqali chiqish tartibi aniqlanadi. Nigora na birinchi, na oxirgi bo'lib chiqishni xohlaydi. Uning istagi ro'yobga chiqish ehtimolini toping.

J: 80%

13. Uy-joy lotereyasida quyidagi yutuqlar o'ynalmoqda:

- Toshkent shahridagi yangi qurilgan uylardan 5 ta kvartira;
- Shahar atrofigidan 2 ta uy;
- «Yangi hayot» hududida qurilish uchun 3 ta yer uchastkasi.

Yutuqlar tasodifiy ravishda taqsimlanadi. Ishtirokchi Dilshodning yutuq qo'lga kiritgani ma'lum. Dilshodning yer uchastkasini yutmagan bo'lish ehtimolini toping.

J: 70%

14. Samarqand universitetida o'tkaziladigan konferensiyada tushgacha 6 ta, tushdan keyin esa 4 ta ma'ruza rejalashtirilgan. Ma'ruzalar tartibi qur'a tashlash orqali aniqlanadi. Professor Hasanovning tushgacha ma'ruza qilishi ehtimolini toping.

J: 60%

15. Buxoroning eski shahrida faoliyat yurituvchi gidlar quyidagicha taqsimlangan: 4 nafari tarixchi, 3 nafari san'atshunos, 3 nafari jurnalist. Turistik firma sayyoh uchun gidni tasodifiy tarzda tayinlaydi. Shahnozaga san'atshunos gid to'g'ri kelish ehtimolini toping.

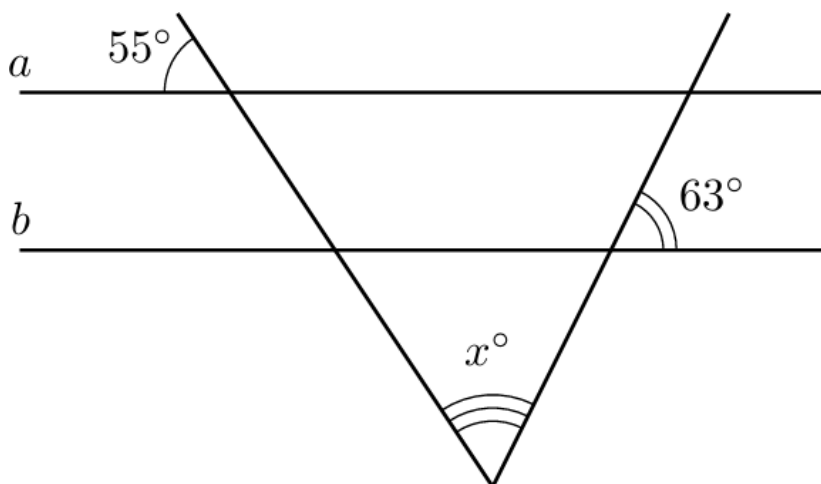
J: 30%

16. Mahallada «Sumalak» bayrami munosabati bilan sovrinlar o'ynalmoqda: 8 ta shirinliklar to'plami, 6 ta «shoyi» dasturxon va 6 ta sopol ko'za. Umidaning bileti yutuqli bo'ldi. Uning shirinliklar to'plami yoki dasturxon yutib olgan bo'lish ehtimolini toping.

J: 70%

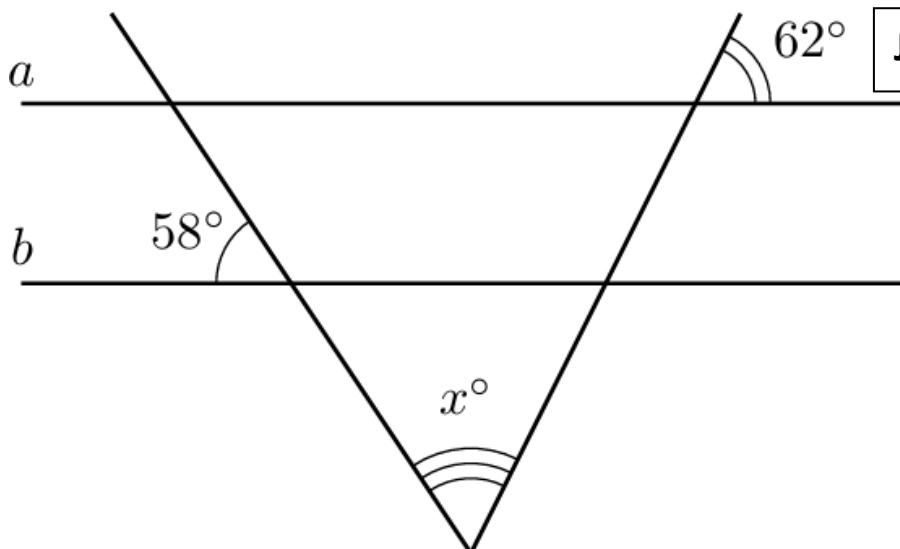
XIII. Burchak

1. Agar a va b to'g'ri chiziqlari parallel bo'lsa, noma'lum x burchakning qiymatini toping.



J: $x=62^\circ$

2. Agar a va b to'g'ri chiziqlari parallel bo'lsa, noma'lum x burchakning qiymatini toping.



J: $x=60^\circ$

3. Parallelogramm burchaklarining biri ikkinchisidan 30° ga katta. Ushbu parallelogrammning katta burchagi qiymatini toping.

J: 105°

4. Parallelogramm burchaklarining biri ikkinchisidan 3 marta katta. Ushbu parallelogrammning kichik burchagi qiymatini toping.

J: 45°

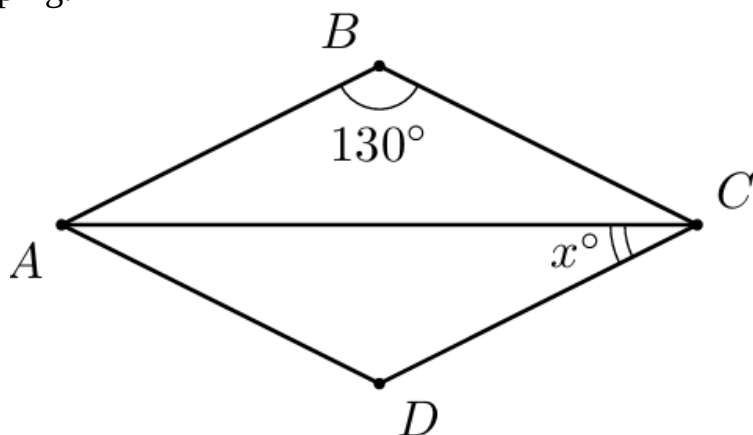
5. Teng yonli trapetsiya burchaklarining biri ikkinchisidan 40° ga katta. Ushbu trapetsiyaning kichik burchagi qiymatini toping.

J: 70°

6. Teng yonli trapetsiya burchaklarining biri ikkinchisidan 4 marta katta. Ushbu trapetsiyaning katta burchagi qiymatini toping.

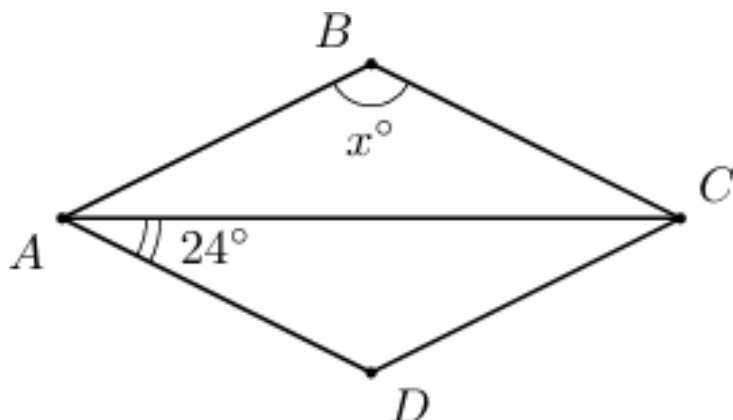
J: 144°

7. Quyidagi rasmda $ABCD$ — romb va $\angle ABC = 130^\circ$ bo'lsa, noma'lum x burchakning qiymatini toping.



J: $x=25^\circ$

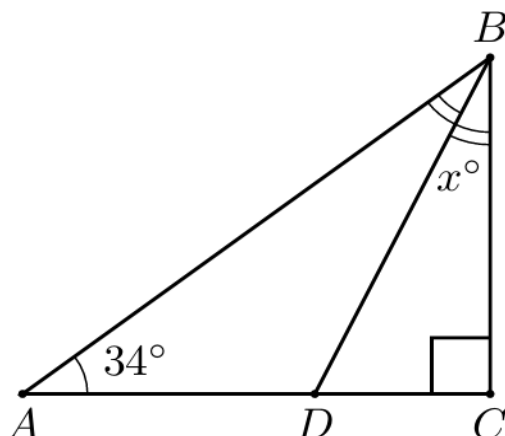
8. Quyidagi rasmda $ABCD$ — romb va $\angle CAD = 24^\circ$ bo'lsa, noma'lum x burchakning qiymatini toping.



J: $x=132^\circ$

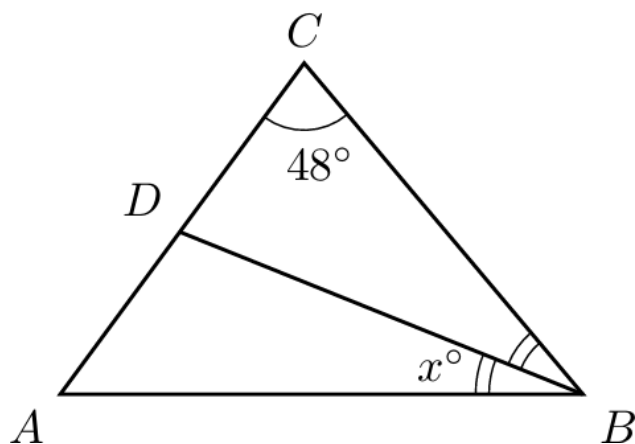
9. Quyidagi rasmda ABC — to'g'ri burchakli uchburchak va BD — ABC burchakning bissektrisasi bo'lsa, noma'lum x burchakning qiymatini toping

J: $x=28^\circ$



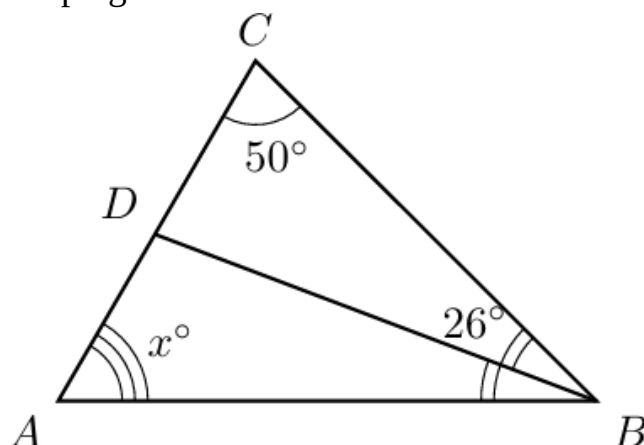
10. Quyidagi rasmda ABC — asosi AB bo'lgan teng yonli uchburchak. Agar BD — ABC burchakning bissektrisasi bo'lsa, noma'lum x burchakning qiymatini toping.

J: $x=33^\circ$



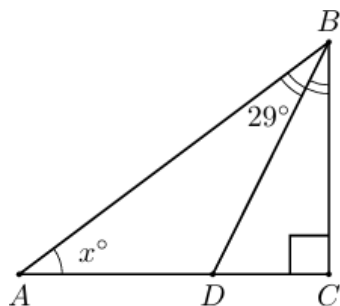
11. Quyidagi rasmda ABC uchburchakda BD bissektrisa o'tkazilgan. Noma'lum x burchakning qiymatini toping.

J: $x=78^\circ$



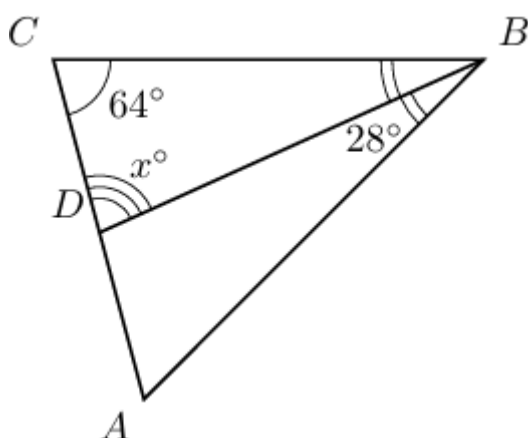
12. Quyidagi rasmda ABC — to'g'ri burchakli uchburchak va BD — ABC burchakning bissektrisasi bo'lsa, noma'lum x burchakning qiymatini toping.

J: $x=32^\circ$



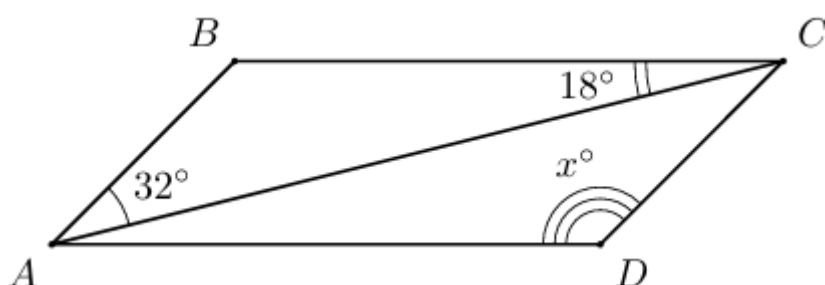
13. Quyidagi rasmda ABC uchburchakda BD bissektrisa o'tkazilgan. Noma'lum x burchakning qiymatini toping.

J: $x=88^\circ$



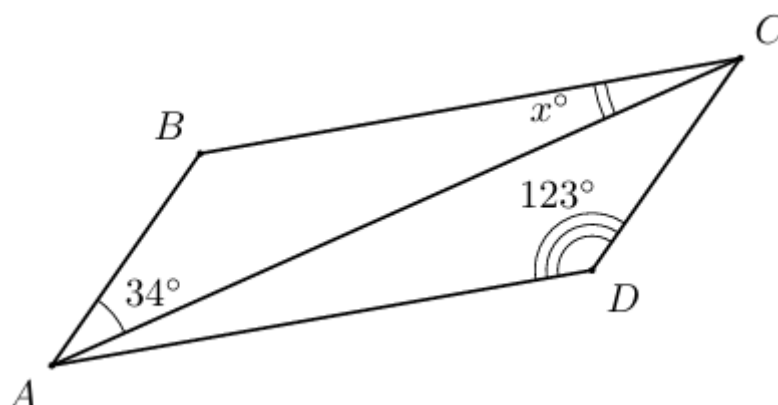
14. Quyidagi rasmda $ABCD$ parallelogramm tasvirlangan. Noma'lum x burchakning qiymatini toping.

J: $x=130^\circ$



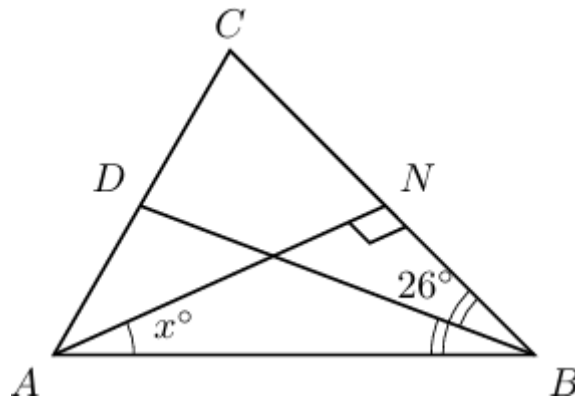
15. Quyidagi rasmda $ABCD$ parallelogramm tasvirlangan. Noma'lum x burchakning qiymatini toping.

J: $x=23^\circ$



16. Quyidagi rasmda ABC uchburchakda BD bissektrisa o'tkazilgan. Noma'lum x burchakning qiymatini toping.

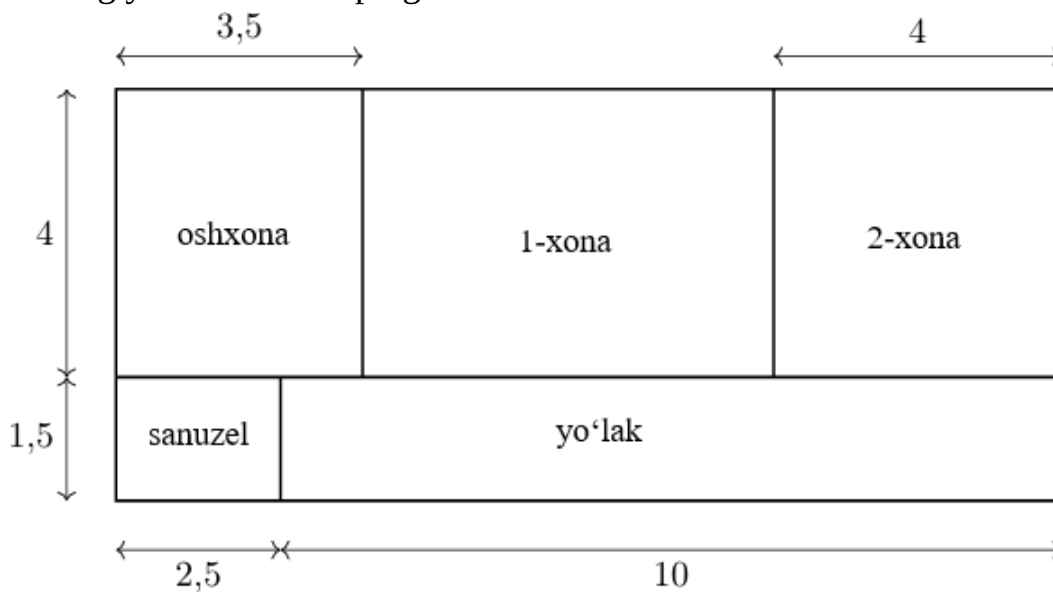
J: $x=38^\circ$



XIV. Amaliy masalalarda yuza va perimetrni topish

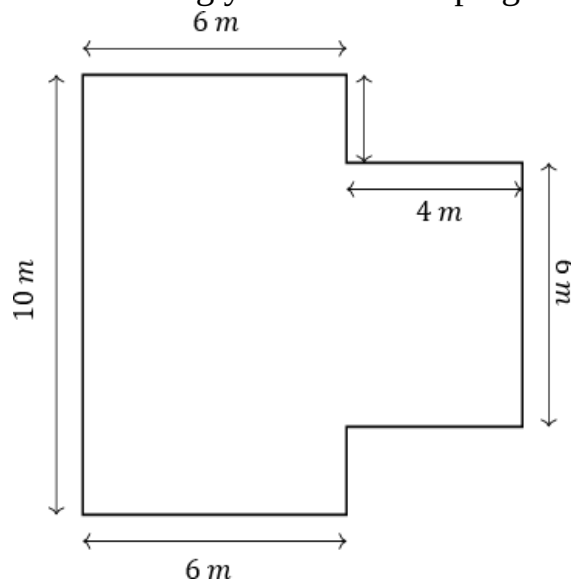
1. Xonadon to'g'ri to'rtburchak shaklida bo'lib, ikki xona, oshxona, yo'lak va sanuzeldan iborat. Oshxonaning o'lchamlari $3,5 \text{ m} \times 4 \text{ m}$, ikkinchi xonaning o'lchamlari $4 \text{ m} \times 4 \text{ m}$, sanuzelning o'lchamlari $1,5 \text{ m} \times 2,5 \text{ m}$, yo'lakning uzunligi esa 10 m ga teng. Birinchi xonaning yuzini m^2 da toping.

J: 20m^2



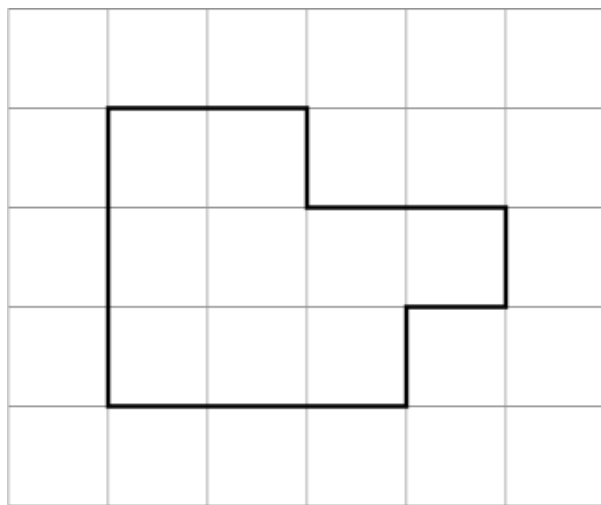
2. Berilgan uy sxemasi asosida uning yuzini m^2 da toping.

J: 84 m^2



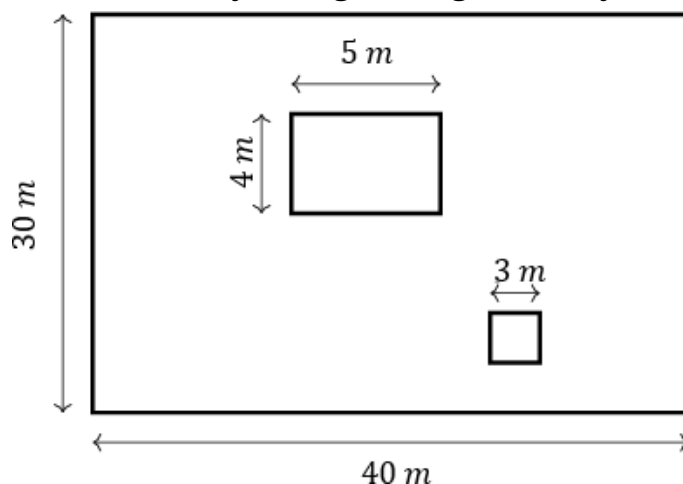
3. Agar katak tomoni 2 m ga teng bo'lsa, rasmda tasvirlangan shaklning yuzini m^2 da toping.

J: $36 m^2$



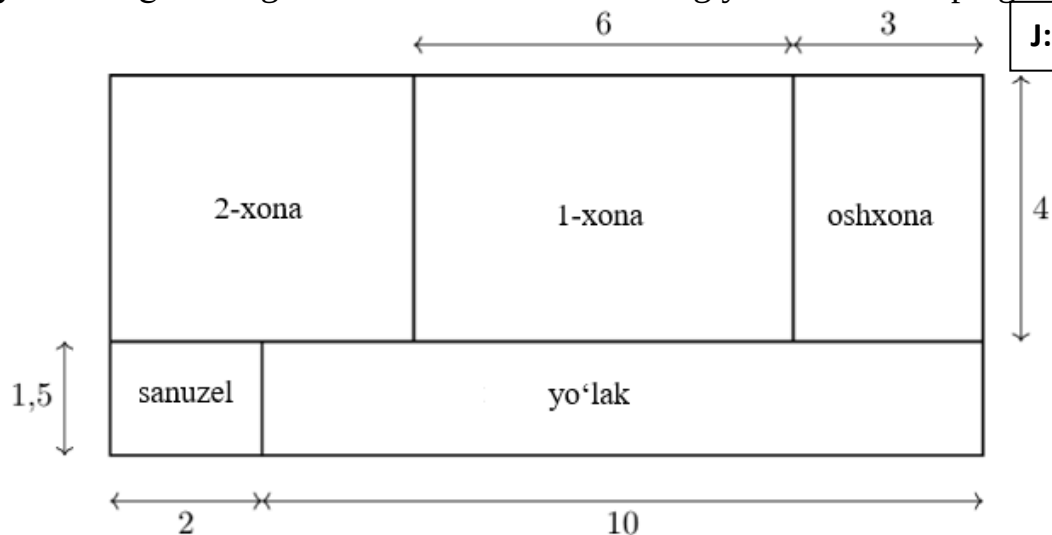
4. O'lchamlari $40 m \times 30 m$ bo'lgan to'g'ri to'rtburchak shaklidagi dala hovlisi berilgan. Hovlida $5 m \times 4 m$ o'lchamdagi omborxona va tomoni 3 m bo'lgan kvadrat shaklidagi ayvon joylashgan. Omborxona va ayvon egallamagan hovli yuzini m^2 da toping.

J: $29 m^2$

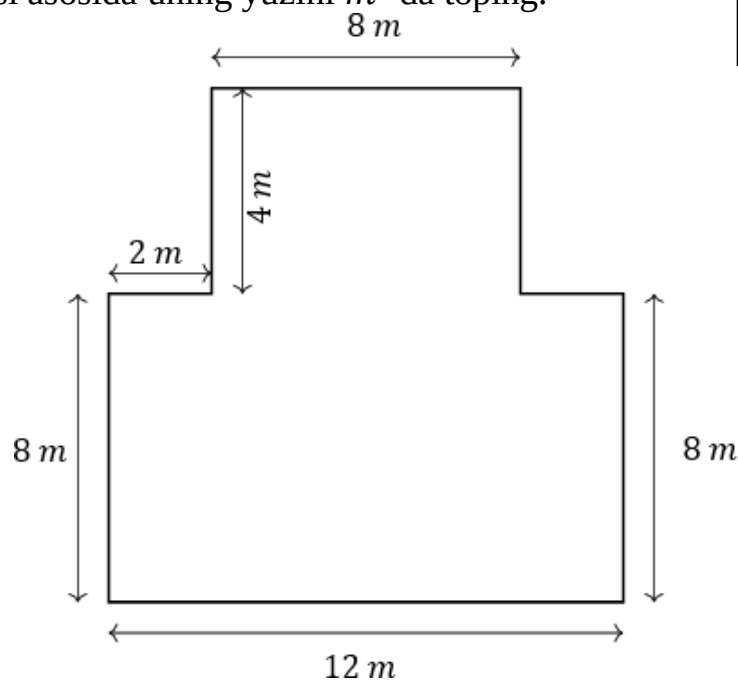


5. Xonadon to'g'ri to'rtburchak shaklida bo'lib, ikki xonadan, oshxonadan, yo'lakdan va sanuzeldan iborat. Oshxona — $3 m \times 4 m$, birinchi xona — $6 m \times 4 m$, sanuzel — $1,5 m \times 2 m$, yo'lakning uzunligi — 10 m. Ikkinchi xonaning yuzini m^2 da toping.

J: $12 m^2$

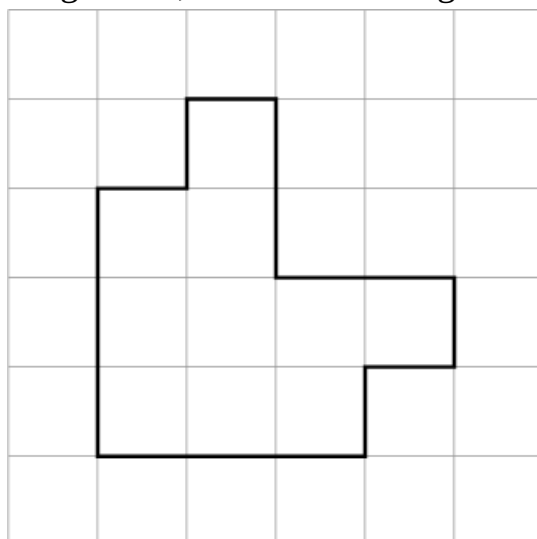


6. Berilgan uy sxemasi asosida uning yuzini m^2 da toping.



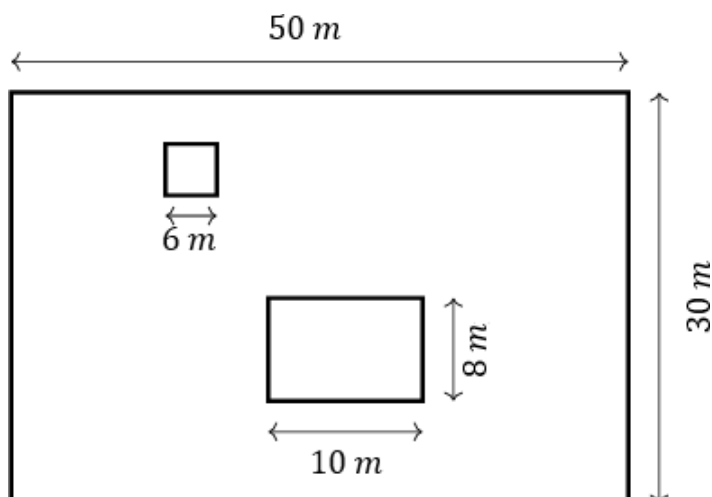
J: $128 m^2$

7. Agar katak tomoni 3 m ga teng bo'lsa, rasmda tasvirlangan shaklning yuzini m^2 da toping.



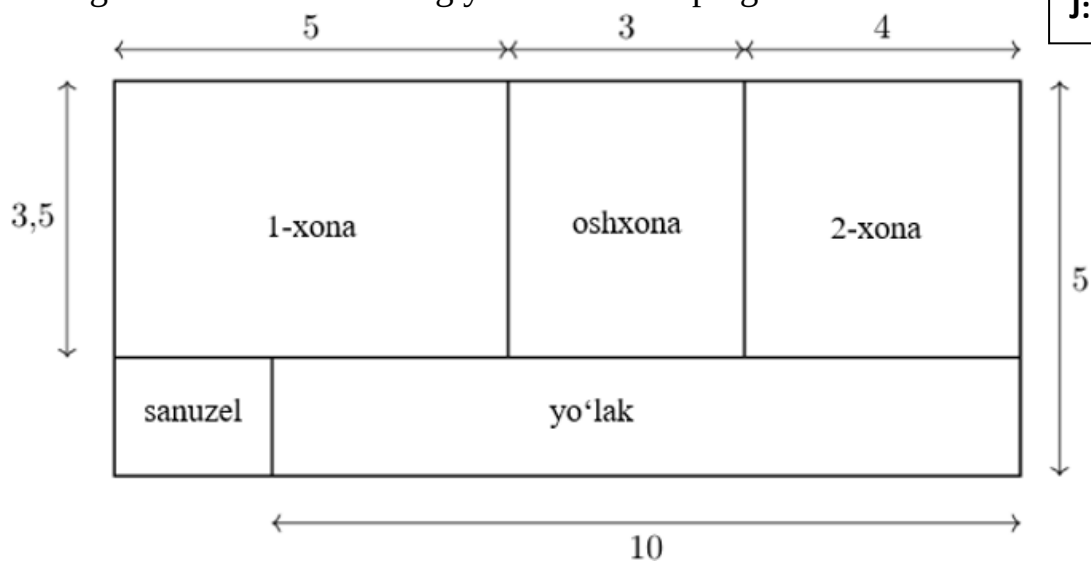
J: $90 m^2$

8. O'lchamlari $50 m \times 30 m$ bo'lgan to'g'ri to'rtburchak shaklidagi dala hovlisi berilgan. Hovlida $10 m \times 8 m$ o'lchamdagi issiqxona va tomoni 6 m bo'lgan kvadrat shaklidagi ayvon joylashgan. Issiqxona va ayvon egallamagan hovli yuzini m^2 da toping.



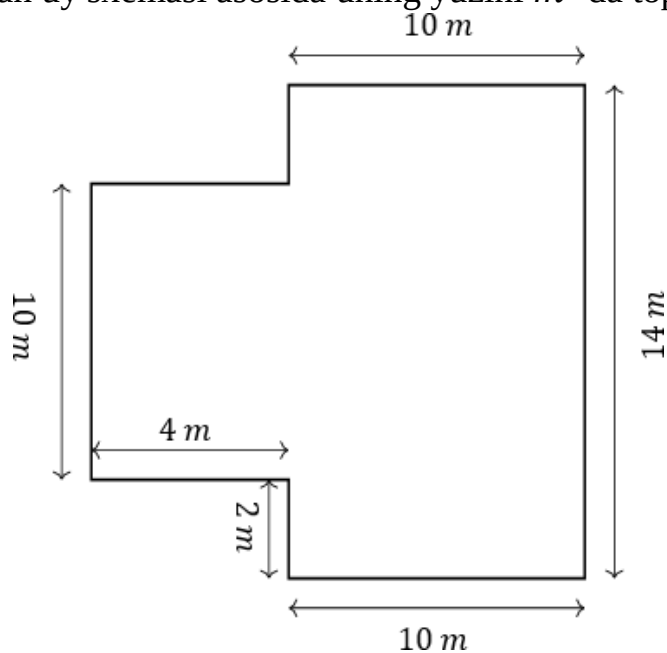
J: $116 m^2$

9. Xonadon to'g'ri to'rtburchak shaklida bo'lib, ikki xonadan, oshxonadan, yo'lakdan va sanuzeldan iborat. Birinchi xona — $3,5 \text{ m} \times 5 \text{ m}$, ikkinchi xona — $4 \text{ m} \times 3,5 \text{ m}$, yo'lakning uzunligi — 10 m . Sanuzelning yuzini m^2 da toping.



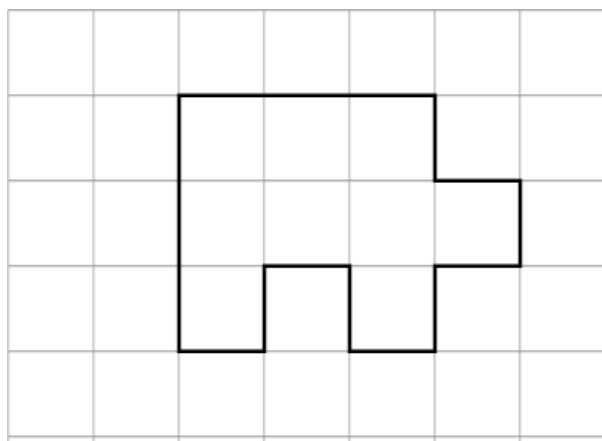
J: 3 m^2

10. Berilgan uy sxemasi asosida uning yuzini m^2 da toping.



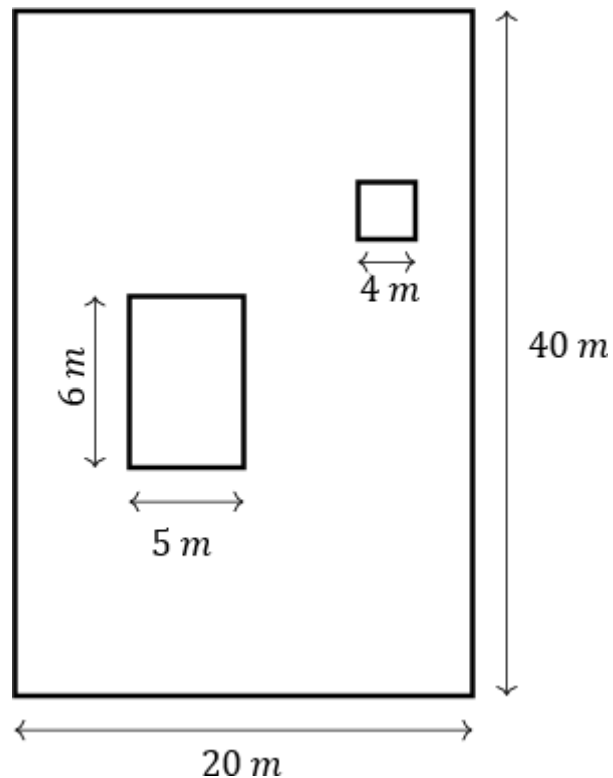
J: 180 m^2

11. Agar katak tomoni 4 m ga teng bo'lsa, rasmda tasvirlangan shaklning yuzini m^2 da toping.



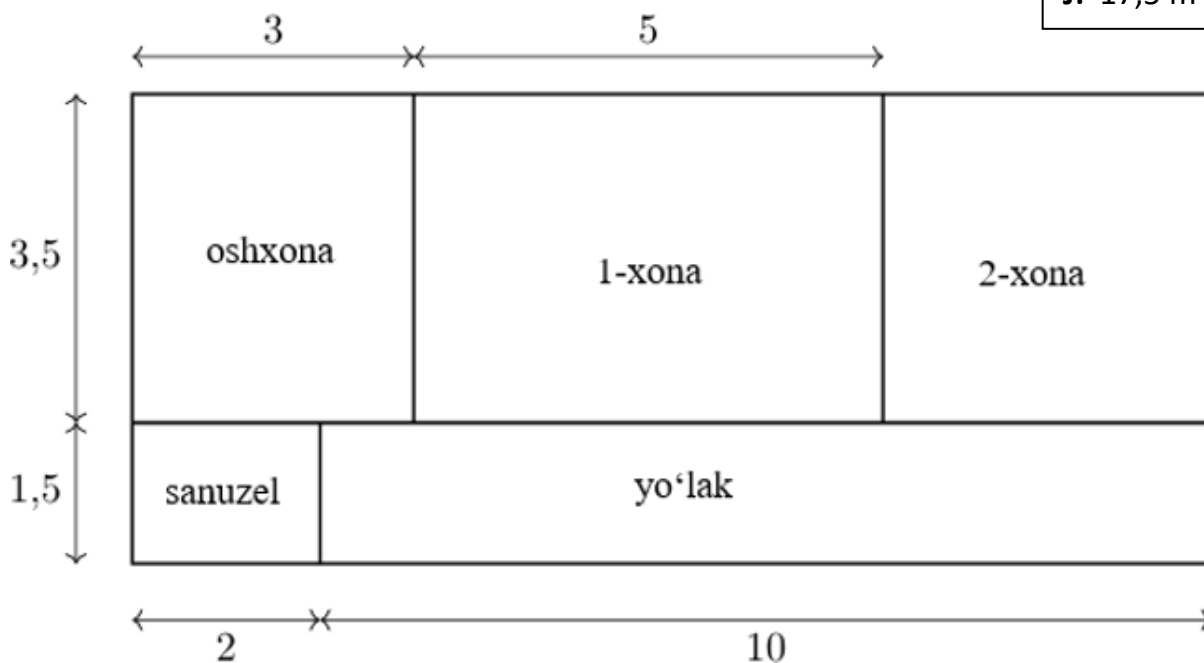
J: 144 m^2

12. O'lchamlari $40\text{ m} \times 20\text{ m}$ bo'lgan to'g'ri to'rtburchak shaklidagi dala hovlisi berilgan. Hovlida $6\text{ m} \times 5\text{ m}$ o'lchamdagi omborxona va tomoni 4 m bo'lgan kvadrat shaklidagi ayvon joylashgan. Omborxona va ayvon egallamagan hovli yuzini m^2 da toping.



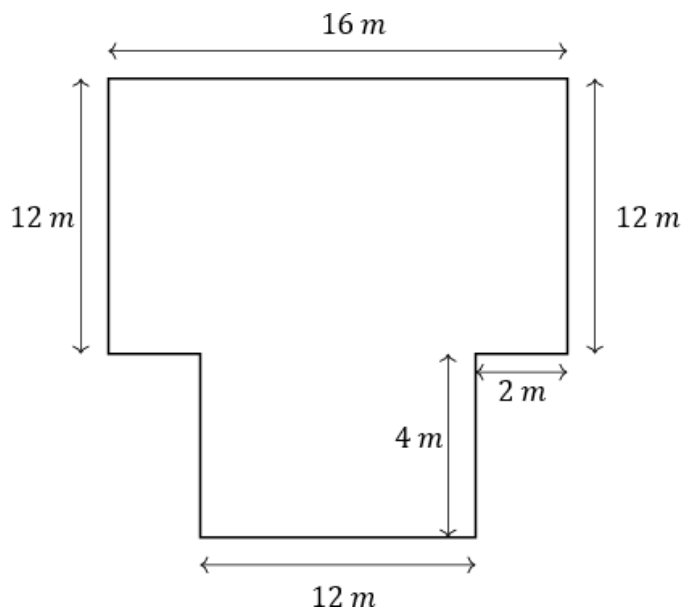
J: 46 m^2

13. Xonadon to'g'ri to'rtburchak shaklida bo'lib, ikki xonadan, oshxonadan, yo'lakdan va sanuzeldan iborat. Oshxona — $3,5\text{ m} \times 3\text{ m}$, birinchi xona — $5\text{ m} \times 3,5\text{ m}$, sanuzel — $1,5\text{ m} \times 2\text{ m}$, yo'lakning uzunligi — 10 m . Ikkinchi xonaning yuzini m^2 da toping.



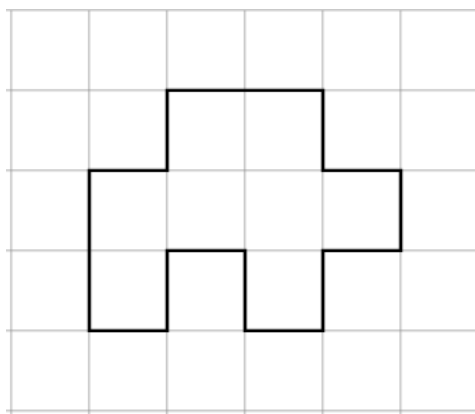
J: $17,5\text{ m}^2$

14. Berilgan uy sxemasi asosida uning yuzini m^2 da toping.



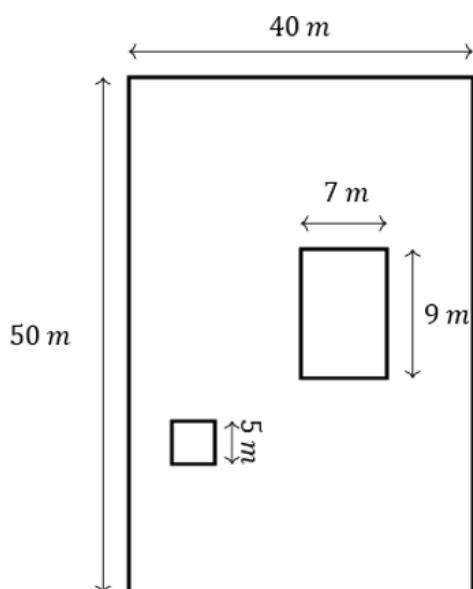
J: $240 m^2$

15. Agar katak tomoni 5 m ga teng bo'lsa, rasmda tasvirlangan shaklning yuzini m^2 da toping.



J: $200 m^2$

16. O'lchamlari $50 m \times 40 m$ bo'lgan to'g'ri to'rtburchak shaklidagi dala hovlisi berilgan. Hovlida $9 m \times 7 m$ o'lchamdagi omborxona va tomoni 5 m bo'lgan kvadrat shaklidagi ayvon joylashgan. Omborxona va ayvon egallamagan hovli yuzini m^2 da toping.



J: $88 m^2$

XV. To'g'ri burchakli uchburchaklar. O'tkir burchakning trigonometrik funksiyalari. Pifagor teoremasi

1. $ABCD$ to'g'ri to'rtburchakda $AB = 6$ cm, $BC = 8$ cm bo'lsa, BAC burchakning sinusini toping. $\underline{J:} \sin A = \frac{4}{5}$
2. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi 15 cm ga, BC kateti esa 9 cm ga teng bo'lsa, B burchakning sinusini toping. $\underline{J:} \sin B = \frac{4}{5}$
3. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi $AB = 26$ cm va $\sin B = \frac{5}{13}$. BC katetning uzunligini toping. $\underline{J:} BC = 24$
4. $ABCD$ to'g'ri to'rtburchakda $AB = 24$ cm, $BC = 7$ cm bo'lsa, $\angle BAC$ burchakning kosinusini toping. $\underline{J:} \cos A = \frac{24}{25}$
5. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi $AB = 25$ cm ga, AC kateti esa 24 cm ga teng bo'lsa, B burchakning kosinusini toping. $\underline{J:} \cos B = \frac{7}{25}$
6. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi $AB = 25$ cm va $\cos B = \frac{7}{25}$. AC katetning uzunligini toping. $\underline{J:} AC = 24$
7. $ABCD$ to'g'ri to'rtburchakda $AB = 12$ cm, $AD = 16$ cm bo'lsa, $\angle ABD$ burchakning sinusini toping. $\underline{J:} \sin B = \frac{4}{5}$
8. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi 30 cm ga, AC kateti esa 24 cm ga teng bo'lsa, A burchakning tangensini toping. $\underline{J:} tg A = \frac{3}{4}$
9. To'g'ri burchagi C bo'lgan ABC uchburchakda $BC = 12$ cm va $tg A = \frac{4}{3}$. AB tomonning uzunligini toping. $\underline{J:} AB = 15$
10. $ABCD$ to'g'ri to'rtburchakda $AC = 25$ cm, $BC = 20$ cm bo'lsa, BAC burchakning kosinusini toping. $\underline{J:} \cos A = \frac{3}{5}$
11. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi 10 cm ga, BC kateti esa 6 cm ga teng bo'lsa, A burchakning kosinusini toping. $\underline{J:} \cos A = \frac{4}{5}$
12. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi $AB = 26$ cm va $\cos A = \frac{12}{13}$. BC tomonning uzunligini toping. $\underline{J:} BC = 10$
13. $ABCD$ to'g'ri to'rtburchakda $AC = 30$ cm, $AB = 18$ cm bo'lsa, BAC burchakning sinusini toping. $\underline{J:} \sin A = \frac{4}{5}$
14. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi 20 cm ga, AC kateti esa 12 cm ga teng bo'lsa, A burchakning sinusini toping. $\underline{J:} \sin A = \frac{4}{5}$

15. To'g'ri burchagi C bo'lgan ABC uchburchakning gipotenuzasi $AB = 34$ cm va $\sin A = \frac{8}{17}$. BC tomonning uzunligini toping. $\underline{J:} \quad BC = 16$

16. $ABCD$ to'g'ri to'rtburchakda $BD = 15$ cm, $CD = 12$ cm bo'lsa, BDC burchakning tangensini toping. $\underline{J:} \quad \tan D = \frac{3}{4}$

XVI. Uchburchaklar va to'rtburchaklarning xossalari

1. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Uchburchakning ichki burchaklari yig'indisi 180° ga teng.
- 2) Parallelogrammning qarama-qarshi tomonlari teng.
- 3) To'g'ri to'rtburchakning diagonalari o'zaro to'g'ri burchak ostida kesishadi.
- 4) Har qanday trapetsiyaning asoslari teng.

1)	2)	3)	4)
T	T	N	N

2. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Teng tomonli uchburchakda barcha burchaklar teng.
- 2) Rombdagi diagonal burchaklarni teng ikkiga bo'ladi.
- 3) To'g'ri to'rtburchakning diagonal uning har qanday tomonidan katta.
- 4) Har qanday trapetsiyaning yon tomonlari teng.

1)	2)	3)	4)
T	T	T	N

3. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) 30° burchakka qarshi yotgan katet gipotenuzaning yarmiga teng.
- 2) Agar parallelogrammning diagonalari teng bo'lsa, u romb bo'ladi.
- 3) Kvadratning diagonalari teng.
- 4) Teng yonli trapetsiyada bir asosga yopishgan burchaklar teng.

1)	2)	3)	4)
T	N	T	T

4. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) To'g'ri burchakli uchburchakning o'tkir burchaklari yig'indisi 90° ga teng.
- 2) Rombdagi barcha burchaklar teng.
- 3) To'g'ri to'rtburchakning diagonalari teng.
- 4) Trapetsiyaning ichki burchaklari yig'indisi 360° ga teng.

1)	2)	3)	4)
T	N	T	T

5. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Tomonlari 1 cm, 2 cm va 4 cm ga teng bo'lgan uchburchak mavjud emas.
- 2) Parallelogrammning qarama-qarshi burchaklari teng.
- 3) Kvadratning diagonallari o'zaro perpendikulyar.
- 4) Trapetsiyaning o'rta chizig'i uning asoslari yig'indisiga teng.

1)	2)	3)	4)
T	T	T	N

6. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) To'g'ri burchakli uchburchakning gipotenuzasi uning katetlari uzunliklari yig'indisidan katta.
- 2) Rombning diagonallari teng.
- 3) To'g'ri to'rtburchakning diagonallari kesishish nuqtasida teng ikkiga bo'linadi.
- 4) To'g'ri burchakli trapetsiyada burchaklardan biri 90° ga teng.

1)	2)	3)	4)
N	N	T	T

7. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) O'tmas burchakli uchburchakda barcha burchaklar o'tmas burchaklar.
- 2) Parallelogrammning ichki burchaklari yig'indisi 360° ga teng.
- 3) Kvadratning barcha burchaklari teng.
- 4) Teng yonli trapetsiyaning asoslari teng.

1)	2)	3)	4)
T	T	T	N

8. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Teng yonli uchburchakning asosiga tushirilgan bissektrisa mediana va balandlik ham bo'ladi.
- 2) Diagonallar parallelogrammni to'rtta teng uchburchakka bo'ladi.
- 3) Kvadratning yuzi uning tomoni kvadratiga teng.
- 4) Trapetsiyada yon tomonlar o'zaro parallel.

1)	2)	3)	4)
T	T	T	N

9. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Uchburchakda katta burchakka qarshi kichik tomon yotadi.
- 2) Parallelogrammda ikkita burchak o'zaro teng.

3) Har qanday kvadrat to'g'ri to'rtburchak hisoblanadi.

4) To'g'ri burchakli uchburchakning barcha burchaklari to'g'ri.

1)	2)	3)	4)
N	T	T	N

10. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

1) Uchburchak medianasi u chiqarilgan burchakni teng ikkiga bo'ladi.

2) Rombning diagonallari kesishish nuqtasida teng ikkiga bo'linadi.

3) Agar parallelogrammning diagonallari teng bo'lsa, u kvadrat bo'ladi.

4) Har qanday trapetsiyaning asoslari o'zaro parallel.

1)	2)	3)	4)
N	T	N	T

11. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

1) Uchburchakning bissektrisasi u tushirilgan tomonni teng ikkiga bo'ladi.

2) Rombning diagonallari uni to'rtta to'g'ri burchakli uchburchakka bo'ladi.

3) Teng yonli uchburchakda barcha burchaklar teng.

4) Teng yonli trapetsiyaning yon tomonlari teng.

1)	2)	3)	4)
N	T	N	T

12. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

1) Har qanday teng tomonli uchburchak teng yonli uchburchakdir.

2) Parallelogrammning barcha burchaklari teng.

3) Kvadratning diagonallari kesishish nuqtasida teng ikkiga bo'linadi.

4) To'g'ri burchakli trapetsiyaning diagonallari teng.

1)	2)	3)	4)
T	N	T	N

13. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

1) Tomonlari 1 cm, 2 cm va 5 cm ga teng bo'lgan uchburchak mavjud.

2) To'g'ri to'rtburchakning diagonallari uni to'rtta teng uchburchakka bo'ladi.

3) Kvadratning diagonallari uning burchaklarini teng ikkiga bo'ladi.

4) Agar uchburchakning barcha burchaklari teng bo'lsa, uning barcha tomonlari ham teng bo'ladi.

1)	2)	3)	4)
N	T	T	T

14. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Parallelogrammning diagonallari uning burchaklarini teng ikkiga bo'ladi.
- 2) Agar parallelogrammning qo'shni tomonlari teng bo'lsa, u romb bo'ladi.
- 3) Teng tomonli uchburchakning medianasi uni ikkita teng uchburchakka bo'ladi.
- 4) Teng yonli trapetsiyaning diagonallari teng.

1)	2)	3)	4)
N	N	T	T

15. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Agar bir uchburchakning burchaklari mos ravishda boshqa uchburchakning burchaklariga teng bo'lsa, bu uchburchaklar teng bo'ladi.
- 2) To'g'ri to'rtburchak parallelogramm hisoblanadi.
- 3) To'g'ri burchakli uchburchakning gipotenuzasi uning katetidan katta.
- 4) Har qanday trapetsiyaning diagonallari teng.

1)	2)	3)	4)
N	T	T	N

16. Quyida keltirilgan tasdiqlarning to'g'ri (T) yoki noto'g'ri (N) ekanini aniqlab, jadvalni to'ldiring.

- 1) Teng yonli uchburchakning asosidagi burchaklar teng.
- 2) Rombning diagonallari o'zaro perpendikulyar.
- 3) To'g'ri to'rtburchakning diagonali uni ikkita teng uchburchakka bo'ladi.
- 4) To'g'ri burchakli trapetsiyada barcha burchaklar to'g'ri.

1)	2)	3)	4)
T	T	T	N

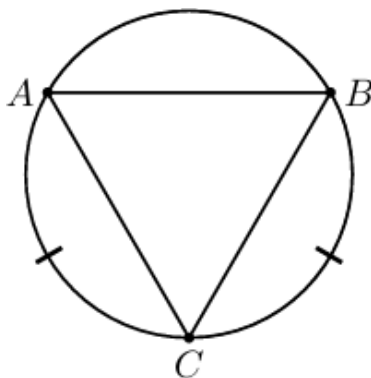
XVII. Ko'pburchaklar yuzlarining formulalari

1. Teng yonli trapetsiyaning asoslari 14 cm va 6 cm ga, yon tomoni esa 5 cm ga teng. Trapetsiyaning yuzini toping. $\underline{\text{J: } S = 30\text{cm}^2}$
2. Teng yonli trapetsiyada asoslar 6 cm va 12 cm ga teng, burchaklardan biri 45° ga teng. Trapetsiyaning yuzini toping. $\underline{\text{J: } S = 27\text{cm}^2}$
3. To'g'ri burchakli trapetsiyaning asoslari 14 cm va 6 cm ga teng, burchaklardan biri 135° ga teng. Trapetsiyaning yuzini toping. $\underline{\text{J: } S = 80\text{cm}^2}$
4. Rombning yuzi 24 cm^2 ga teng, uning diagonallaridan biri 12 cm. Rombning tomonini toping. $\underline{\text{J: } a = 2\sqrt{5}}$
5. To'g'ri burchakli trapetsiyaning asoslari 16 cm va 6 cm ga teng, katta yon tomoni 26 cm. Trapetsiyaning yuzini toping. $\underline{\text{J: } S = 264\text{cm}^2}$

6. Rombning tomoni 13 cm ga teng, diagonallaridan biri 10 cm. Rombning yuzini toping.
J: $S = 120cm^2$
7. Teng yonli uchburchakning yon tomonlari 13 cm ga, asosi 10 cm ga teng.
Uchburchakning yuzini toping. J: $S = 60cm^2$
8. To'g'ri burchakli uchburchakning yuzi $30 cm^2$ ga teng, katetlaridan biri 10 cm.
Uchburchakning gipotenuzasini toping. J: $\sqrt{136}$
9. To'g'ri to'rtburchakning bir tomoni 8 cm, diagonali 10 cm ga teng. Diagonallar kesishishidan hosil bo'ladigan uchburchaklarning yuzini toping. J: $S = 12cm^2$
10. To'g'ri burchakli trapetsiyaning yon tomonlari 6 cm va 10 cm ga, kichik asosi 8 cm ga teng. Trapetsiyaning yuzini toping. J: $S = 72cm^2$
11. Trapetsiyaning o'rta chizig'i 12 cm ga teng. Yon tomonlardan biri 8 cm bo'lib, asos bilan 30° burchak hosil qiladi. Trapetsiyaning yuzini toping. J: $S = 48cm^2$
12. Teng yonli uchburchakning yuzi $30 cm^2$ ga, asosi 10 cm ga teng. Uchburchakning yon tomoni uzunligini toping. J: $\sqrt{61}$
13. To'g'ri burchakli uchburchakning gipotenuzasi 20 cm, katetlaridan biri 16 cm. Ushbu uchburchakning o'rta chiziqlari hosil qilgan uchburchakning yuzini toping.
J: $S = 24cm^2$
14. To'g'ri burchakli trapetsiyada kichik asos va kichik yon tomoni 4 cm ga teng.
Burchaklardan biri 135° ga teng. Trapetsiyaning yuzini toping. J: $S = 24cm^2$
15. Teng yonli trapetsiyaning yuzi $180 cm^2$ ga teng, asoslari 6 cm va 24 cm.
Trapetsiyaning yon tomonini toping. J: 15
16. To'g'ri to'rtburchakning diagonali 8 cm ga teng va u to'g'ri burchakni 1:2 nisbatda bo'ladi. To'g'ri to'rtburchakning yuzini toping. J: $S = 8\sqrt{3}cm^2$

XVIII Aylana. Urinma. Aylanadagi burchaklar

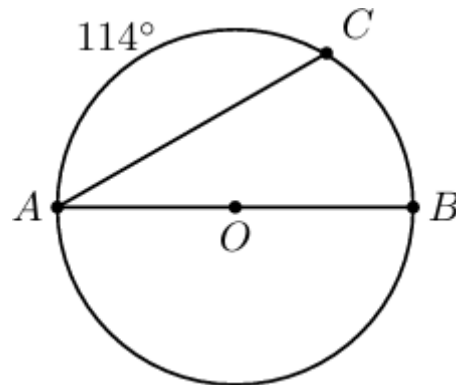
1. Quyidagi rasmda ACB ichki burchagi 58° ga teng bo'lsa, BAC burchakning qiymatini toping.



J: $\angle BAC = 61^\circ$

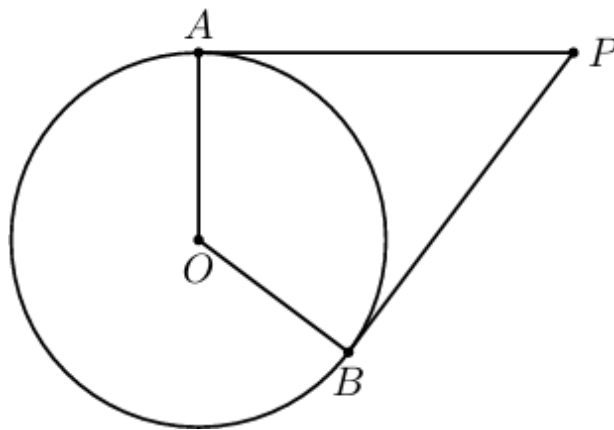
2. Quyidagi rasmda AB — aylana diametri va AC yoy 114° ga teng bo'lsa, CAB burchakning qiymatini toping.

J: $\angle CAB = 33^\circ$

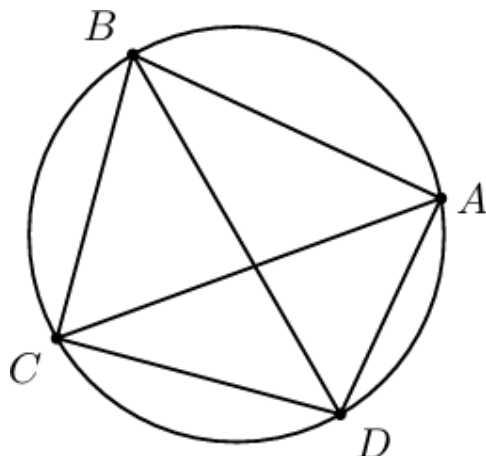


3. Quyidagi rasmda P nuqtadan markazi O nuqtada bo'lgan aylanaga ikki urinma o'tkazilgan. Agar $\angle APB = 38^\circ$ bo'lsa, $\angle AOB$ ning qiymatini toping.

J: $\angle AOB = 142^\circ$

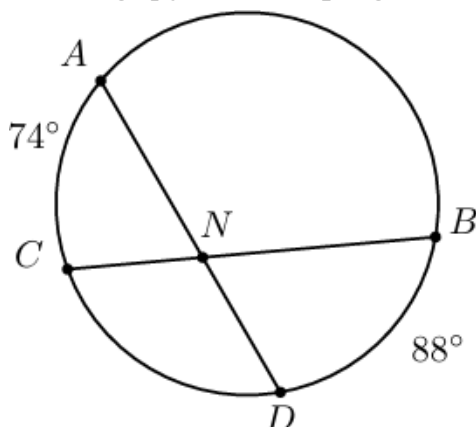


4. Quyidagi rasmda $\angle CDB = 44^\circ$ va $\angle CAD = 42^\circ$ bo'lsa, $\angle BAD$ ning qiymatini toping.



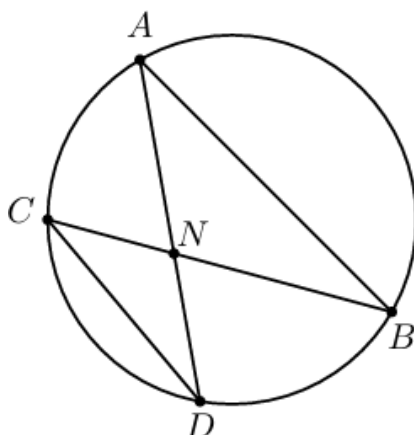
J: $\angle BAD = 86^\circ$

5. Quyidagi rasmda AD va BC vatarlari N nuqtada kesishadi. Agar AC yoy 74° ga, DB yoy esa 88° ga teng bo'lsa, $\angle BND$ ning qiymatini toping.



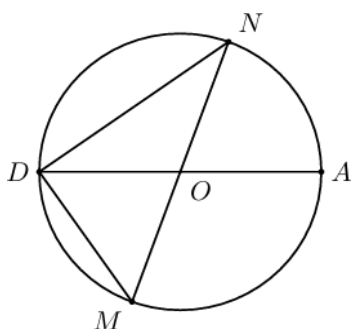
J: $\angle BND = 81^\circ$

6. Quyidagi rasmda $\angle DAB = 60^\circ$ va $\angle CDA = 20^\circ$ bo'lsa, $\angle ANB$ ning qiymatini toping.



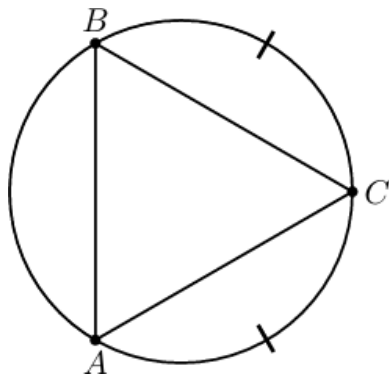
J: $\angle ANB = 100^\circ$

7. Quyidagi rasmda $\angle NDA = 32^\circ$. Agar DA diametr bo'lsa, $\angle NMD$ ning qiymatini toping.



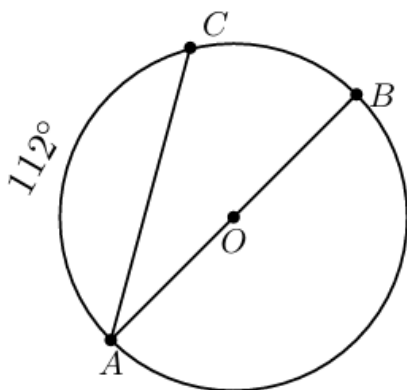
J: $\angle NMD = 58^\circ$

8. Quyidagi rasmda ichki $\angle ACB = 54^\circ$ bo'lsa, $\angle BAC$ ning qiymatini toping.



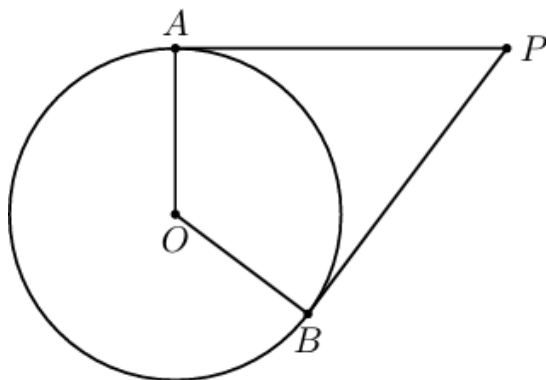
J: $\angle BAC = 63^\circ$

9. Quyidagi rasmda AB — aylana diametri va AC yoy 112° ga teng bo'lsa, $\angle CAB$ ning qiymatini toping.



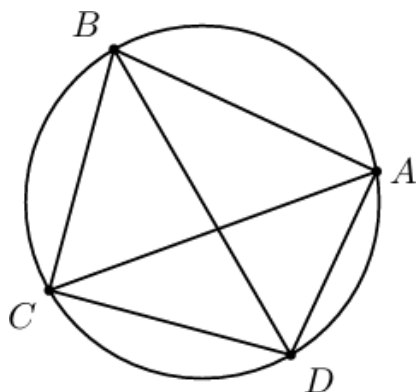
J: $\angle CAB = 34^\circ$

10. Quyidagi rasmda P nuqtadan markazi O nuqtada bo'lgan aylanaga ikki urinma o'tkazilgan. Agar $\angle APB = 113^\circ$ bo'lsa, $\angle AOB$ ning qiymatini toping.



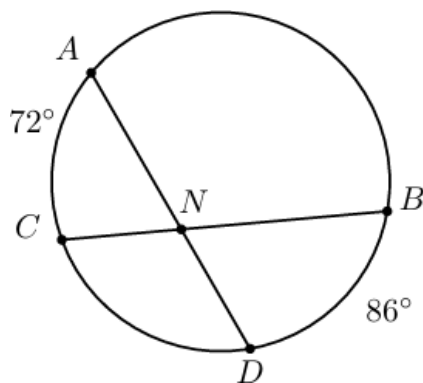
J: $\angle AOB = 67^\circ$

11. Quyidagi rasmda $\angle CDB = 42^\circ$ va $\angle CAD = 45^\circ$ bo'lsa, $\angle BAD$ ning qiymatini toping.



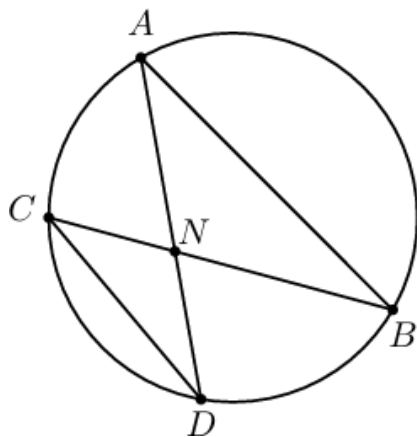
J: $\angle BAD = 97^\circ$

12. Quyidagi rasmda AD va BC vatarlari N nuqtada kesishadi. Agar AC yoy 72° ga, DB yoy esa 86° ga teng bo'lsa, $\angle BND$ ning qiymatini toping.



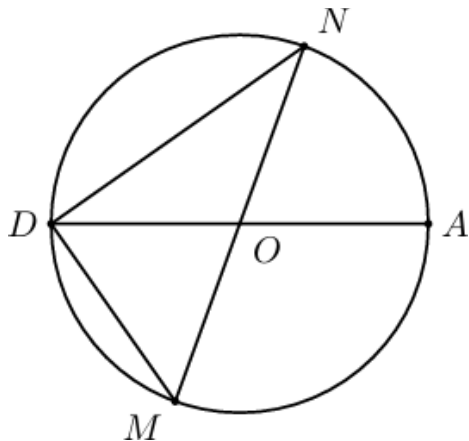
J: $\angle BMD = 79^\circ$

13. Quyidagi rasmda $\angle DAB = 42^\circ$ va $\angle CDA = 27^\circ$ bo'lsa, $\angle ANB$ ning qiymatini toping.



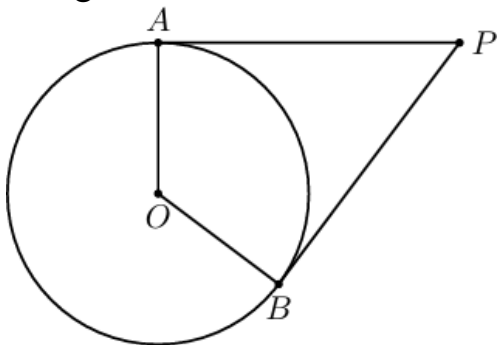
J: $\angle ANB = 42^\circ$

14. Quyidagi rasmda $\angle NDA = 36^\circ$. Agar DA diametr bo'lsa, $\angle NMD$ ning qiymatini toping.



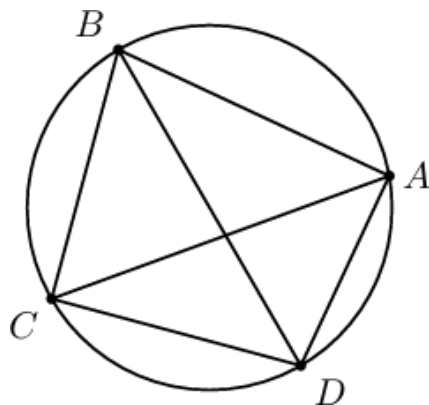
J: $\angle NMD = 54^\circ$

15. Quyidagi rasmda P nuqtadan markazi O nuqtada bo'lgan aylanaga ikki urinma o'tkazilgan. Agar $\angle APB = 46^\circ$ bo'lsa, $\angle AOB$ ning qiymatini toping.



J: $\angle AOB = 134^\circ$

16. Quyidagi rasmda $\angle CDB = 46^\circ$ va $\angle CAD = 41^\circ$ bo'lsa, $\angle BAD$ ning qiymatini toping.



J: $\angle BAD = 87^\circ$

XIX Muntazam ko'pburchaklar va ularga ichki va tashqi chizilgan aylana

- Aylanaga tashqi teng tomonli uchburchak va ichki kvadrat chizilgan. Agar uchburchak tomoni $8\sqrt{60}$ cm bo'lsa, kvadrat tomoni uzunligini toping.
J: 8 cm
- Aylanaga tashqi kvadrat va ichki teng tomonli uchburchak chizilgan. Agar kvadrat tomoni uzunligi $6\sqrt{3}$ cm bo'lsa, uchburchak tomoni uzunligini toping.
J: 9 cm
- Aylanaga muntazam oltiburchak ichki chizilgan, muntazam uchburchak tashqi chizilgan. Agar oltiburchak tomoni $3\sqrt{3}$ cm bo'lsa, uchburchak tomonini toping.
J: 18 cm
- Aylanaga kvadrat ichki chizilgan, muntazam oltiburchak tashqi chizilgan. Agar kvadrat tomoni $12\sqrt{6}$ cm bo'lsa, oltiburchak tomonini toping.
J: 24 cm
- Muntazam uchburchakka tashqi chizilgan aylananing uzunligi $12\sqrt{3}\pi$ cm ga teng. Uchburchakning tomonini toping.
J: 18 cm
- Kvadratga ichki chizilgan doiraning yuzi 8π cm^2 ga teng. Doira markazidan kvadrat uchigacha bo'lgan masofani toping.
J: 4 cm
- Muntazam oltiburchakka ichki chizilgan aylananing uzunligi $6\sqrt{3}\pi$ cm ga teng. Oltiburchakning tomonini toping.
J: 6 cm
- Muntazam uchburchakka tashqi chizilgan doiraning yuzi 27π cm^2 ga teng. Uchburchakning tomonini toping.
J: 9 cm
- Kvadratga tashqi chizilgan aylananing uzunligi $2\sqrt{2}\pi$ cm ga teng. Aylana markazidan kvadrat tomonigacha bo'lgan masofani toping.
J: 1 cm

10. Muntazam uchburchakka ichki chizilgan doiraning yuzi $36\pi \text{ cm}^2$ ga teng. Doira markazidan uchburchak uchigacha bo'lgan masofani toping.
J: 12 cm
11. Muntazam oltiburchakning tomoni $4\sqrt{3}$ cm ga teng. Oltiburchakka tashqi chizilgan aylana markazidan oltiburchak tomonigacha bo'lgan masofani toping.
J: 6 cm
12. Muntazam oltiburchakka tashqi chizilgan doiraning yuzi $64\pi \text{ cm}^2$ ga teng. Oltiburchakning tomonini toping.
J: 8 cm
13. Muntazam uchburchakning tomoni $12\sqrt{3}$ cm ga teng. Uchburchakka ichki chizilgan aylana markazidan uchburchak uchigacha bo'lgan masofani toping.
J: 12 cm
14. Aylanaga tomoni $8\sqrt{2}$ cm bo'lgan kvadrat va muntazam oltiburchak ichki chizilgan. Oltiburchakning tomonini toping.
J: 8 cm
15. Aylanaga tomoni $4\sqrt{6}$ cm bo'lgan muntazam uchburchak va kvadrat ichki chizilgan. Kvadratning tomonini toping.
J: 8 cm
16. Aylanaga tomoni $10\sqrt{3}$ cm bo'lgan muntazam uchburchak va muntazam oltiburchak ichki chizilgan. Oltiburchakning tomonini toping.
J: 10 cm

XX. O'xshashlik. Sinuslar va kosinuslar teoremlari tatbiqlari

1. Asoslari 6 cm va 12 cm, balandligi $3\sqrt{7}$ cm bo'lgan teng yonli trapetsiya berilgan. Trapetsiyaning diagonali diagonallar kesishish nuqtasida qanday uzunlikdagi kesmalarga ajraladi?
J: 4 cm, 8 cm
2. Katetlari 9 cm va 12 cm bo'lgan to'g'ri burchakli uchburchakning gipotenuzasida M nuqta olingan. Agar M nuqtadan kichik burchak uchigacha bo'lgan masofa 5 cm ga teng bo'lsa, shu nuqtadan to'g'ri burchak uchigacha bo'lgan masofani toping.
J: $\sqrt{73}$
3. Radiuslari 4 cm va 6 cm bo'lgan ikki aylana bir xil burchakka ichki chizilgan va ular o'zaro kesishishmaydi. Aylanalar markazlari orasidagi masofa 13 cm ga teng bo'lsa, burchak uchidan kichik aylana markazigacha bo'lgan masofani toping.
J: 26 cm
4. ABC uchburchakda $AB = 6 \text{ cm}$, $BC = 7 \text{ cm}$, $AC = 8 \text{ cm}$. N nuqta — AB tomonning o'rtasi, K nuqta esa BC tomonida joylashgan bo'lib, $BK = 2 \text{ cm}$. NK kesmaning uzunligini toping.

J: $NK = \sqrt{10}$

5. Radiuslari 6 cm va 8 cm bo'lgan ikki aylana bir xil burchakka ichki chizilgan va ular o'zaro urinadi. Burchak uchidan kichik aylana markazigacha bo'lgan masofani toping.

J: 42 cm

6. Asoslari $BC = 5$ cm va $AD = 20$ cm bo'lgan $BCDA$ trapetsiyada AC diagonalini o'tkazilgan va $\angle BAC = \angle CDA$ ekani ma'lum. Agar O nuqta — diagonalalar kesishish nuqtasi bo'lsa, AO kesmaning uzunligini toping.

J: $AO = 8$ cm

7. Yon tomoni 30 cm bo'lgan teng yonli uchburchak radiusi 25 cm bo'lgan aylanaga ichki chizilgan. Uchburchakning asosi uzunligini toping.

J: 48 cm

8. $ABCD$ to'g'ri to'rtburchakda AC diagonalini o'tkazilgan. Diagonalda A uchidan 5 cm masofada N nuqta olingan. Agar $AD = 9$ cm, $DC = 12$ cm bo'lsa, DN masofani toping.

J: $DN = 2\sqrt{13}$

9. Uchburchakning balandligi 12 cm bo'lib, u tushirilgan tomonni 16 cm va 9 cm kesmalarga ajratadi. Ushbu uchburchakka tashqi chizilgan aylananing radiusini toping.

J: 12,5 cm

10. $ABCD$ to'g'ri to'rtburchakda AC diagonalini 13 cm ga, $AB = 5$ cm ga teng. BC tomonida B uchidan 4 cm masofada N nuqta olingan. Agar AN to'g'ri chiziq CD tomonni uning davomidagi K nuqtada kesib o'tsa, KD kesmaning uzunligini toping.

J: $KD = 5$ cm

11. $ABCD$ rombda $AC = 24$ cm, $BD = 18$ cm. BC tomonida $BE = 5$ cm bo'ladigan E nuqta olingan. AE kesmaning uzunligini toping.

J: $2\sqrt{73}$

12. $ABCD$ to'g'ri burchakli trapetsiyaning asoslari $BC = 6$ cm ga va $AD = 8$ cm ga, katta yon tomoni $CD = 10$ cm ga teng. A nuqtadan CD tomonining o'rtasigacha bo'lgan masofani toping.

J: $\sqrt{73}$ cm

13. Asosi 12 cm va yon tomoni 10 cm bo'lgan teng yonli uchburchakka tashqi chizilgan aylananing radiusini toping.

J: 6,25 cm

14. $ABCD$ parallelogrammda $AB = 3\sqrt{3}$ cm, $AD = 6$ cm, $\angle ADC = 30^\circ$. AD tomonida $DE = 2$ cm bo'ladigan E nuqta olingan. E nuqtadan AB ga parallel chiziq o'tkazilib, u BD diagonalni F nuqtada kesib o'tadi. DF kesmaning uzunligini toping.

J: $DF\sqrt{13}$

15. Yon tomoni 15 cm va asosi $AC = 6$ cm bo'lgan teng yonli ABC uchburchakda BC tomonida $BK = 5$ cm bo'ladigan K nuqta olingan. AK kesmaning uzunligini toping.

J: $AK=\sqrt{7}$

16. $ABCD$ to'g'ri to'rtburchakda E nuqta AD tomonida joylashgan. $AB = 6$ cm, $AD = 8$ cm, $AE = 2$ cm. BE kesma AC diagonalni K nuqtada kesib o'tadi. AK kesmaning uzunligini toping.

J: $AK=2$ cm

